Mapping Language Models to Grounded Conceptual Spaces



Roma Patel and Ellie Pavlick

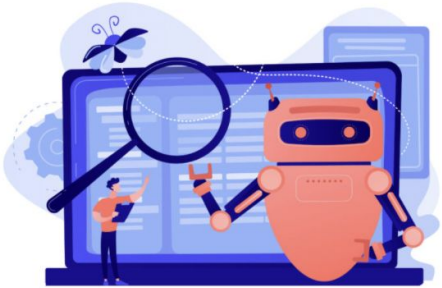
IST Seminar.

Language models in the last few years..

May 25, 2021

Researchers Evaluate Neural Language Models, Find XLNet Excellent

Oliver Peckham



Lexical substitution is essentially the process that a thesaurus helps humans to perform: replacing words in a sentence without changing the meaning. Now, researchers from the Skolkovo Institute of Science and Technology in Moscow – Skoltech, for short – have completed a groundbreaking, large-scale study to examine how the most advanced neural language models perform when handling lexical substitution tasks.

While simple on paper, humans, of course

Language models in the last few years..

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Better Language Models

MIT News

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Teaching language models grammar really does make them smarter

Language Processing, Machine Learning

How Large Language Models Will Transform Science, Society, and AI

Scholars in computer science, linguistics, and philosophy explore the pains and promises of GPT-3.

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Artificial intelligence / Machine learning

Language models like GPT-3 could herald a new type of search engine

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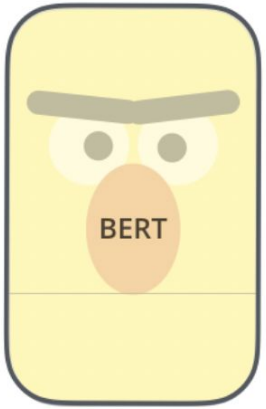
How Large Language Models Will Transform Science, Society, and AI

Scholars in computer science, linguistics, and philosophy explore the pains and promises of GPT-3.

Both classes of language models have their merits



Generative language models *(GPT-x) that are trained to generate text left->right with self-attention*

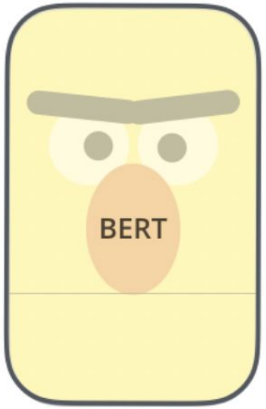


Discriminative language models *(BERT-style) that are trained with a masked language modelling objective*

Both classes of language models have their merits



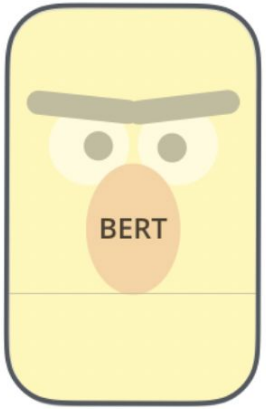
Generative language models *(GPT-x) that are trained to generate text left->right with self-attention*



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Impressive performance of pre-trained models

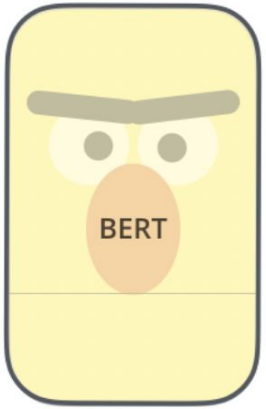




Performance on NLP tasks is significantly better than most previous models and methods

Impressive performance of pre-trained models





Can generate fluent, coherent and human-like text (QA, summarisation)

Title: United Methodists Agree to Historic Split Subtitle: Those who oppose gay marriage will form their own denomination Article: **After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings.**

Language Models are Few-Shot Learners. Brown et. al., NeuRIPs 2020.

Can few-shot learn NLU tasks: entailment, sentiment, grammatical corrections

Poor English input: The patient was died.
Good English output: The patient died.

Poor English input: We think that Leslie likes ourselves.
Good English output: We think that Leslie likes us.

Poor English input: Janet broke Bill on the finger.
Good English output: Janet broke Bill's finger.

Poor English input: Mary arranged for, in St. Louis, John to rent a house cheap.
Good English output: Mary arranged for John to rent a house in St. Louis.

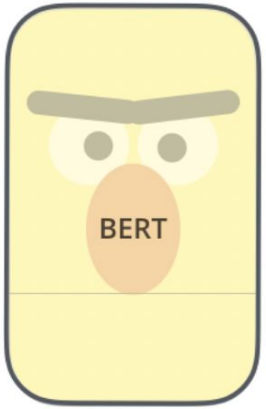
Context →	The bet, which won him dinner for four, was regarding the existence and mass of the top quark, an elementary particle discovered in 1995. question: The Top Quark is the last of six flavors of quarks predicted by the standard model theory of particle physics. True or False? answer:
Target Completion →	False

Figure G.31: Formatted dataset example for RTE

GLUE: A Multi-task Benchmark and Analysis Platform for Natural Language Understanding. Wang et. al., ICLR 2019.

Impressive performance of pre-trained models





Can reconstruct WordNet structure and learn all word relations!

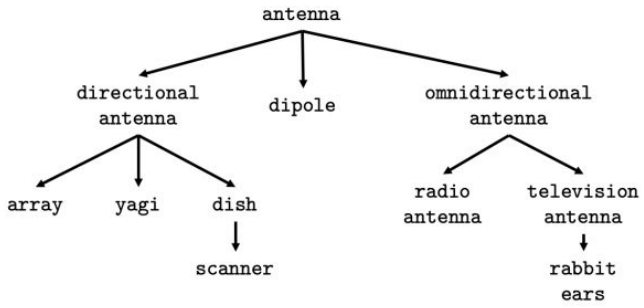
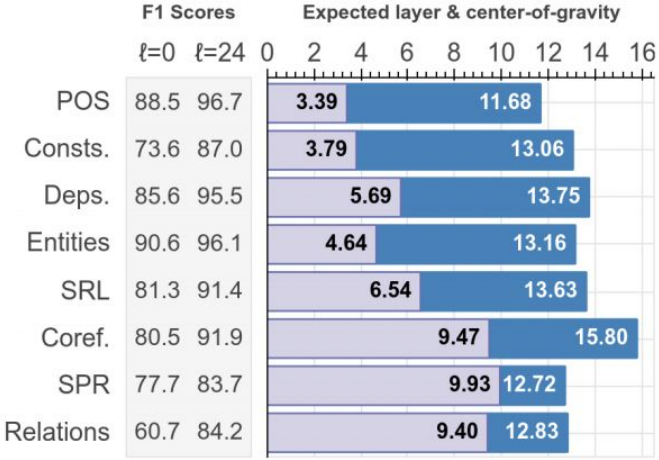


Figure 1: *An example subtree from the* WORDNET *hierarchy.*

Constructing Taxonomies from Pretrained Language Models. Chen, Lin and Klein. NAACL 2021.

Can solve most classical NLP tasks!



BERT Rediscovers the Classical NLP Pipeline. Tenney, Das and Pavlick. ACL 2019.

Models, by definition, are not “grounded”

“We argue that the language modeling task, because it only uses form as training data, cannot in principle lead to learning of meaning”.

Bender and Koller. ACL 2020.

Models, by definition, are not “grounded”

“We argue that the language modeling task, because it only uses form as training data, cannot in principle lead to learning of meaning”.

Bender and Koller. ACL 2020.



How can a model know what the words blue and green mean by only seeing the word forms?

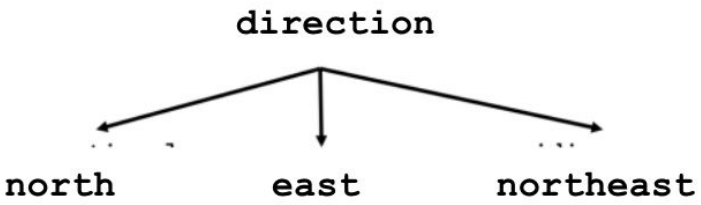


blue



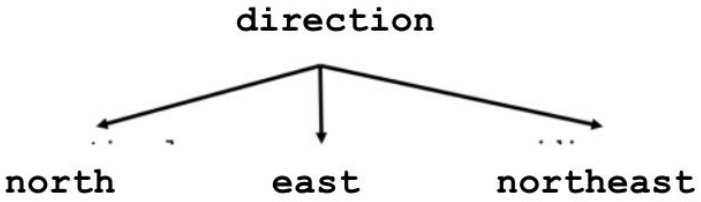
green

They do seem to know the relations between word forms





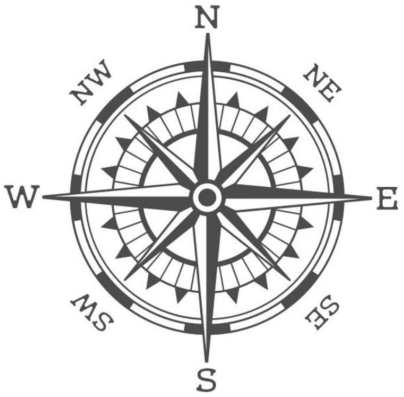
They do seem to know the relations between word forms



How do we take the relational structure that language models seem to know, and allow them to learn the grounded meaning of concepts?



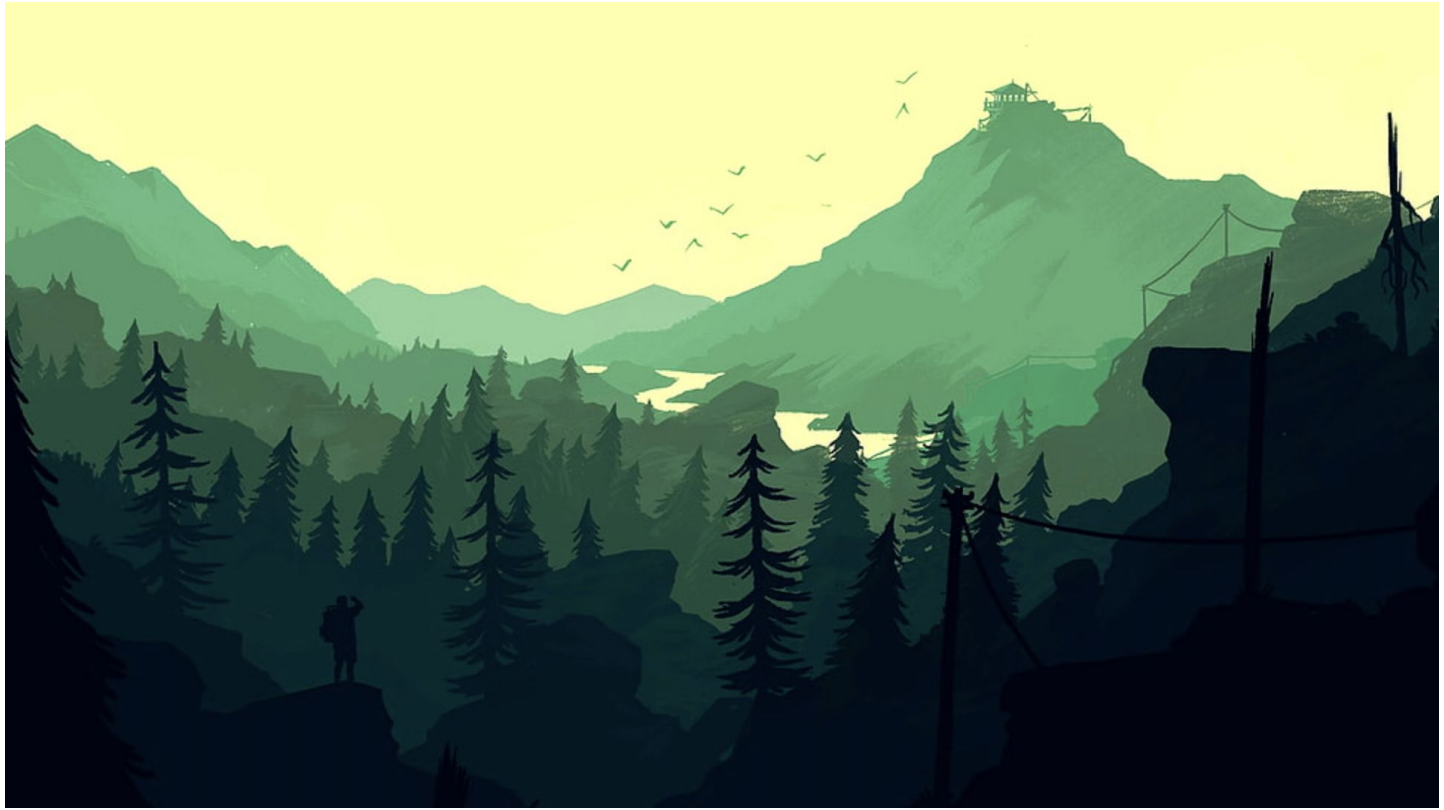
north →

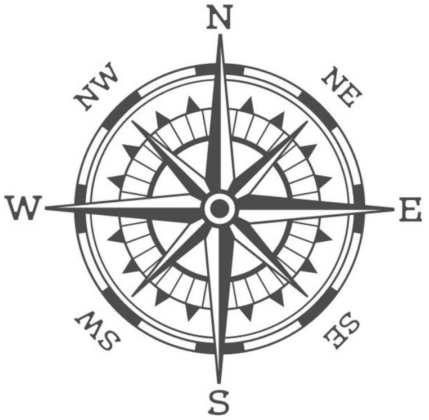


Consider a scenario where you are lost in a forest...

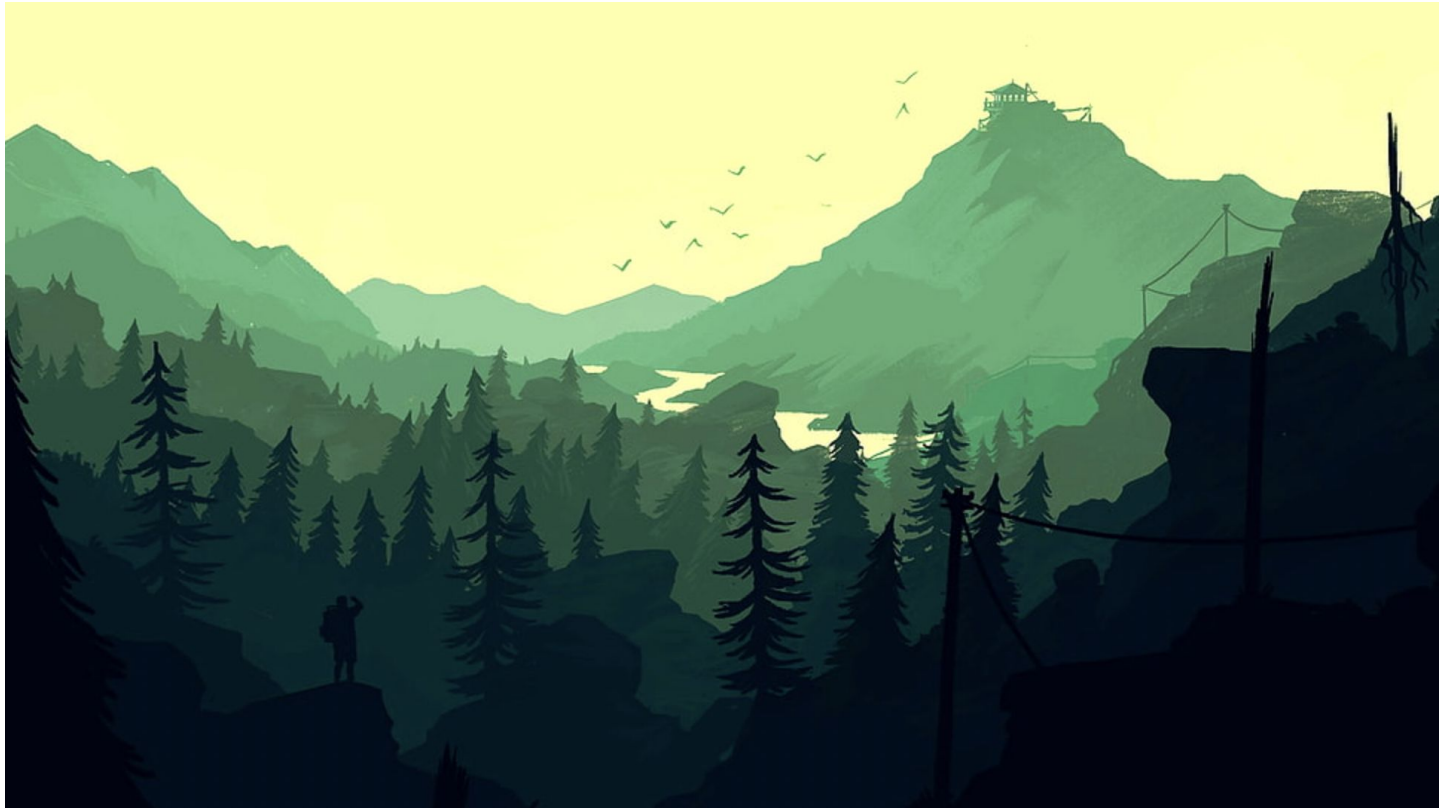


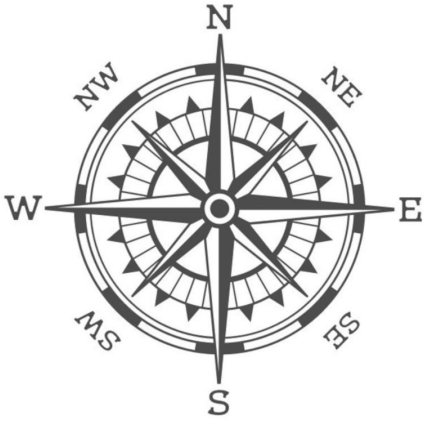
Consider a scenario where you are lost in a forest...





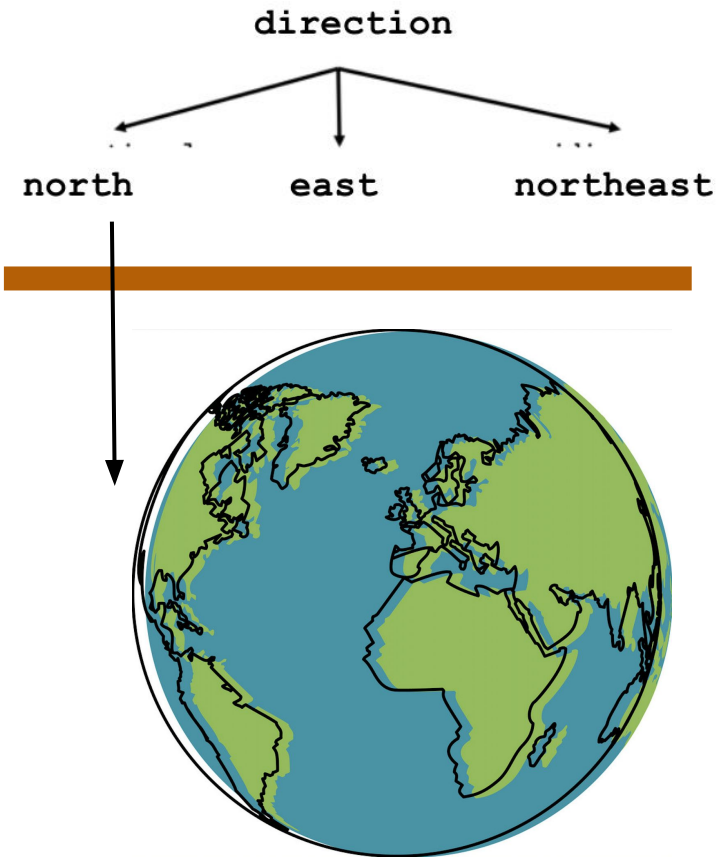
Consider a scenario where you are lost in a forest...



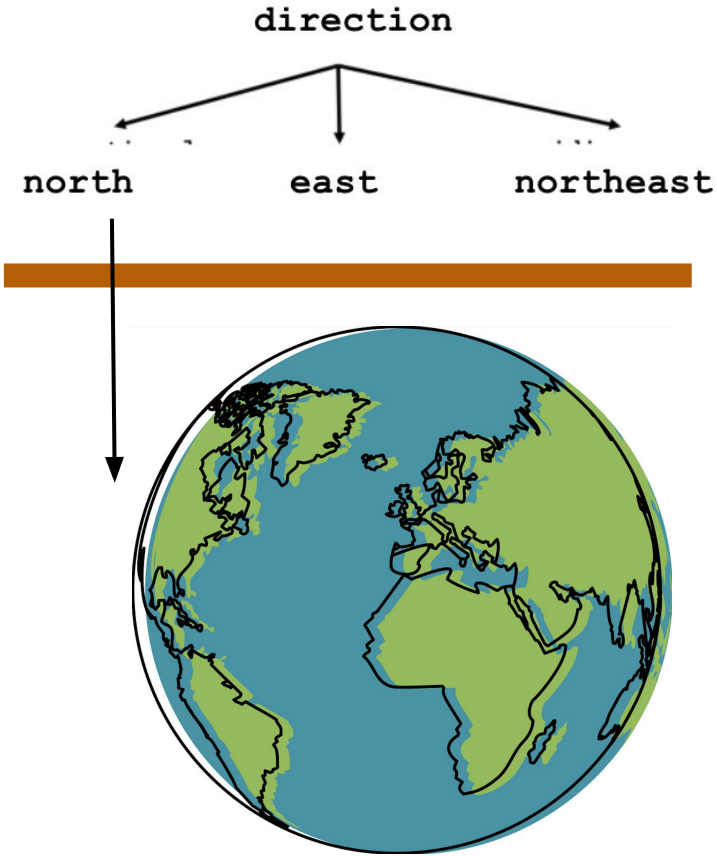


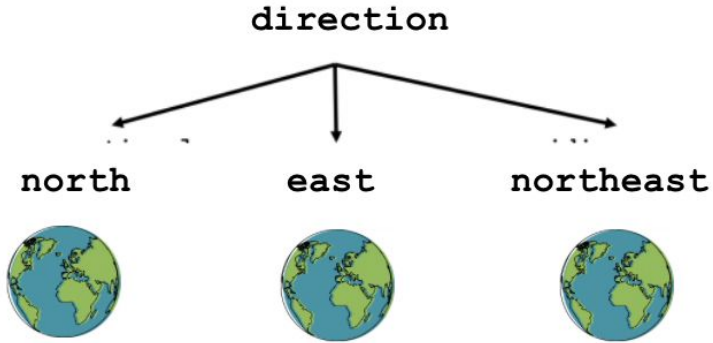
If you know where "north" is, and know the relation between "north, south, east, west", you now know how to orient yourself!

A toe-hold in groundedness



A toe-hold in groundedness

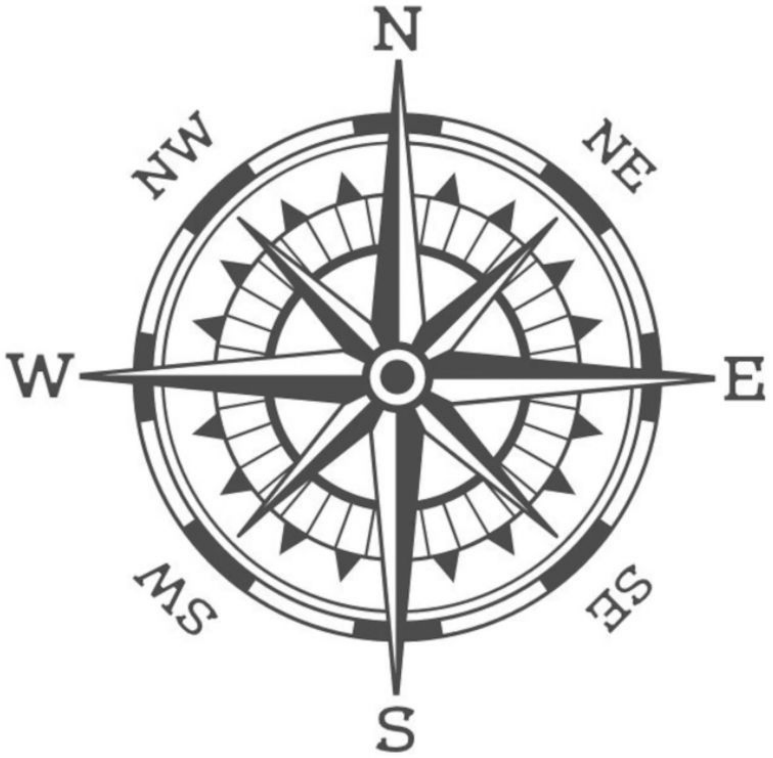




*If we teach models how to ground **some** concepts, can they use the structure of language to extrapolate meaning to **all** related concepts?*

Grounded Concept Categories

Grounded Concept Categories



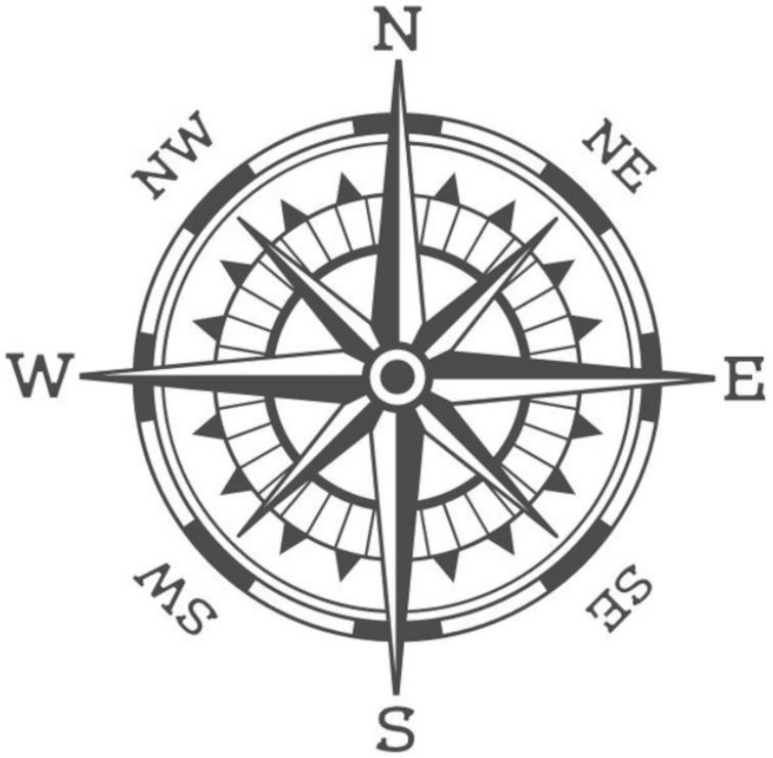
Concept: Spatial directions

If an LM knows all spatial words e.g., *left*, *right*, *up*...

If an LM knows how they relate to each other on a spectrum.

If we show it what it means to be *left*, can it learn what *right* means?

Grounded Concept Categories



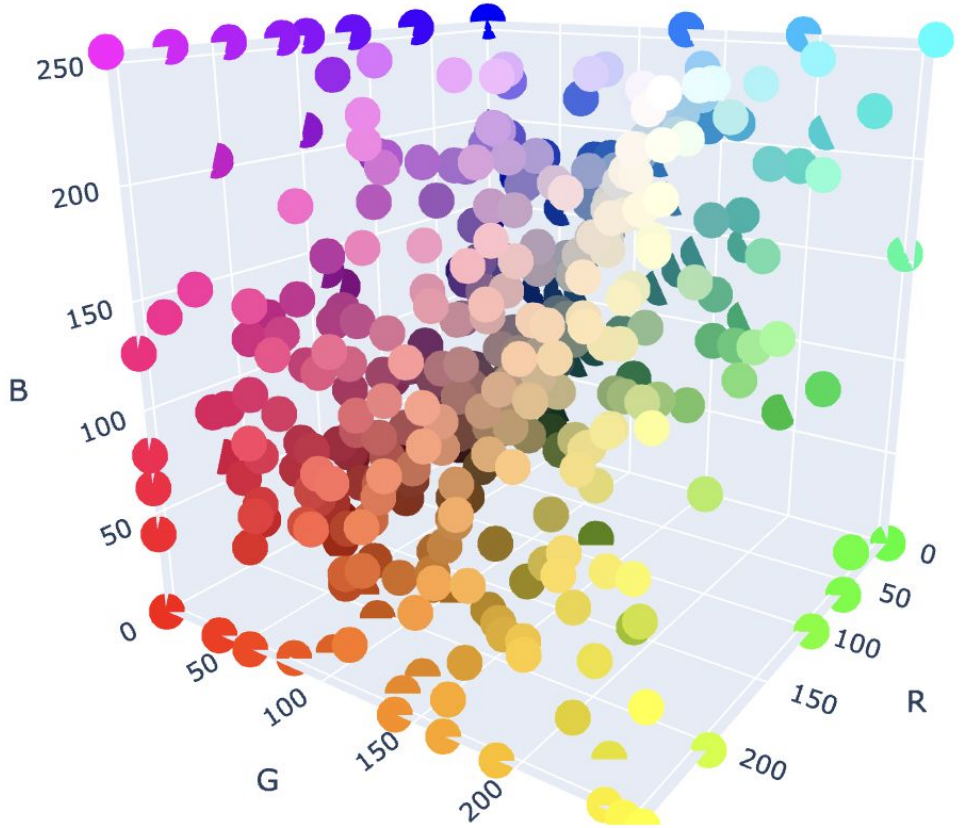
Concept: Cardinal directions

If an LM knows all spatial words e.g., *north, south, east, west...*

If an LM knows how they relate to each other on a spectrum.

If we show it what it means to be *north*, can it learn what *south, east, west* mean?

Grounded Concept Categories



Concept: Colour

If an LM knows all colour words e.g., *lavender, purple, pink, violet, blue...*

If an LM knows which colours are ***warm colours*** vs. ***cool colours***..

If we show it what it means to be a ***cool colour*** by showing it ***blue***, can it now know the grounded meaning of every colour on the spectrum?

What models might be able to quickly learn concepts?

What models might be able to quickly learn concepts?

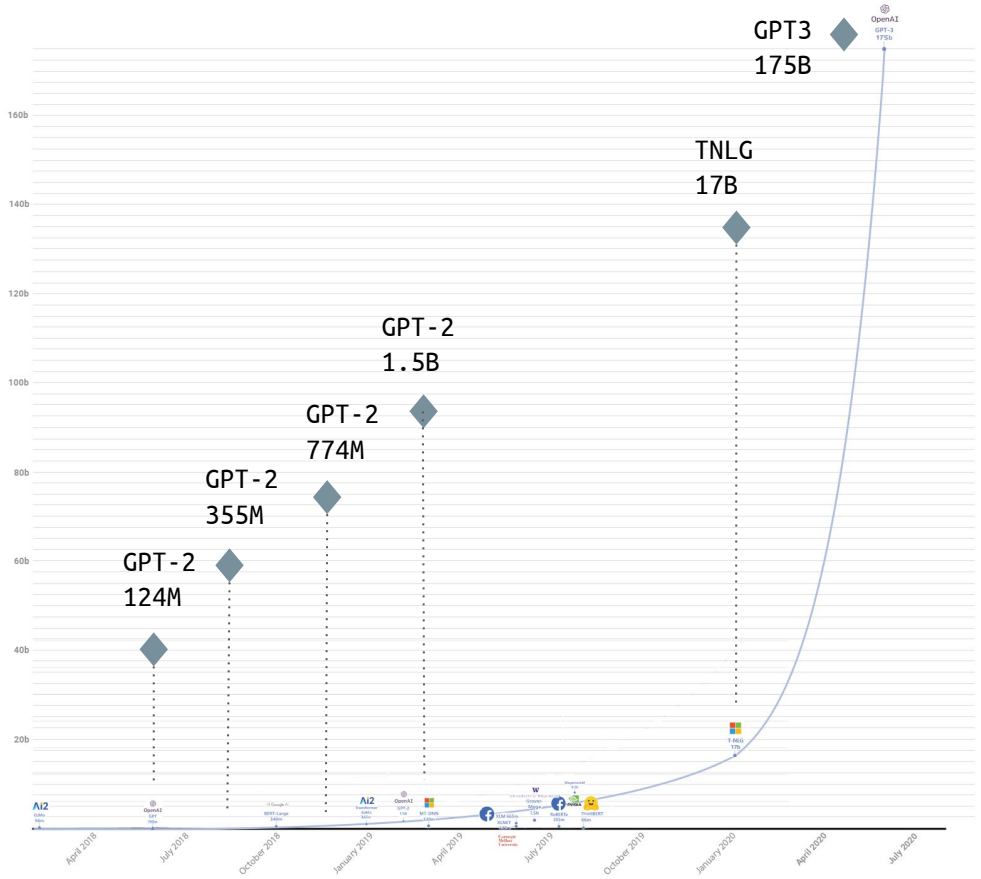
Models that can adapt quickly to new samples (grounded concepts) while preserving pretrained information (relations between concept forms)

GPT-like generative language models

Our models:

GPT-2 (124 M)
GPT-2 (355 M)
GPT-2 (774 M)
GPT-2 (1.5 B)
TNLG (17 B)
GPT-3 (175 B)

All language models trained to generate words sequentially



Few-shot prompts for models

Few-shot prompts for models

Handcrafted prompts for few-shot tuning

```
Translate French to English:
Mon corps est un transformateur de soi,
mais aussi un  transformateur pour cette
cire de langage.
```

Figure 1. Prompt to teach models to translate from French to English.

Few-shot prompts for models

Handcrafted prompts for few-shot tuning

```
Translate French to English:
Mon corps est un transformateur de soi,
mais aussi un  transformateur pour cette
cire de langage.
```

Figure 1. Prompt to teach models to translate from French to English.

Different ways you could few-shot prompt the model for the same answer

```
Question:
Answer:
```

```
Q:
A:
```

```
World:
Description:
```

```
World:
The description is:
```

Few-shot prompts for models

Prompt style 1

```
RGB: (0, 0, 255)
Answer: blue

RGB: (255, 0, 0)
Answer: red

RGB: (255, 255, 0)
Answer: yellow

RGB: (255, 165, 0)
Answer: orange

RGB: (143, 0, 255)
Answer: violet

RGB: (0, 255, 0)
Answer: green

RGB: (29, 172, 214)
Answer: bright cerulean

RGB: (223, 0, 255)
Answer: phlox

RGB: (0, 65, 106)
Answer: dark imperial blue

RGB: (111, 0, 255)
Answer: electric indigo

RGB: (239, 222, 205)
Answer:
```

Prompt style 2

```
RGB: (0, 0, 255)
Colour name: blue

RGB: (255, 0, 0)
Colour name: red

RGB: (255, 255, 0)
Colour name: yellow

RGB: (255, 165, 0)
Colour name: orange

RGB: (143, 0, 255)
Colour name: violet

RGB: (0, 255, 0)
Colour name: green

RGB: (229, 170, 112)
Colour name: fawn

RGB: (217, 144, 88)
Colour name: persian orange

RGB: (247, 127, 190)
Colour name: persian pink

RGB: (0, 206, 209)
Colour name: dark turquoise

RGB: (239, 222, 205)
Colour name:
```

Prompt style 3

```
RGB: (0, 0, 255)
The colour name is: blue

RGB: (255, 0, 0)
The colour name is: red

RGB: (255, 255, 0)
The colour name is: yellow

RGB: (255, 165, 0)
The colour name is: orange

RGB: (143, 0, 255)
The colour name is: violet

RGB: (0, 255, 0)
The colour name is: green

RGB: (233, 116, 81)
The colour name is: burnt sienna

RGB: (255, 0, 127)
The colour name is: rose

RGB: (192, 192, 192)
The colour name is: silver

RGB: (231, 254, 255)
The colour name is: bubbles

RGB: (239, 222, 205)
The colour name is:
```

Example answers

```
rusty orange
dark taupe
light amber
orange
wine
dark blue
dark red
light apricot
.
.
.
```

Few-shot prompts for models

Prompt style 1

```
Array
[[0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Position:  right

Array
[[0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 1. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]]
Position:  right

Array
[[0. 1. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Position:  left

Array
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 1.]
 [0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
Position:  right
```

Prompt style 2

```
Array
[[0. 0. 0. 0. 1. 0. 0.]]
Direction:  right

Array
[[1. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]]
Direction:  left

Array
[[0. 1. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Direction:  left

Array
[[0. 0. 0. 1. 0.]
 [0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
Direction:  right
```

Prompt style 3

```
Array
[[0. 0.]
 [0. 1.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]]
The direction is:  right

Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right

Array
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]
 [1. 0. 0. 0. 0.]]
The direction is:  left
```

Example answers

left

right

.

.

.

How do we ensure models aren’t memorising grounded concepts?

How do we ensure models aren't memorising grounded concepts?



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Matrix (mathematics)

From Wikipedia, the free encyclopedia

For other uses, see Matrix.
"Matrix theory" redirects here. For the physics topic, see Matrix theory (physics).

In mathematics, a **matrix** (plural **matrices**) is a rectangular array or table of numbers, symbols, or expressions, arranged in rows and columns, which is used to represent a mathematical object or a property of such an object. For example,

$$\begin{bmatrix} 1 & 9 & -13 \\ 20 & 5 & -6 \end{bmatrix}$$

is a matrix with two rows and three columns; one say often a "two by three matrix", a "2×3-matrix", or a matrix of dimension 2×3.

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How do we ensure models aren't memorising grounded concepts?

Description [edit]

A matrix of the form

$$L = \begin{bmatrix} \ell_{1,1} & & & & 0 \\ \ell_{2,1} & \ell_{2,2} & & & \\ \ell_{3,1} & \ell_{3,2} & \ddots & & \\ \vdots & \vdots & \ddots & \ddots & \\ \ell_{n,1} & \ell_{n,2} & \dots & \ell_{n,n-1} & \ell_{n,n} \end{bmatrix}$$

is called a **lower triangular matrix** or **left triangular matrix**, and analogously a matrix of the form

$$U = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} & \dots & u_{1,n} \\ & u_{2,2} & u_{2,3} & \dots & u_{2,n} \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & u_{n-1,n} \\ 0 & & & & u_{n,n} \end{bmatrix}$$

is called an **upper triangular matrix** or **right triangular matrix**. A lower or left triangular matrix is commonly denoted with the variable *L*, and an upper or right triangular matrix is commonly denoted with the variable *U* or *R*.

Examples [edit]

This matrix

$$\begin{bmatrix} 1 & 4 & 1 \\ 0 & 6 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

is upper triangular and this matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 8 & 0 \\ 4 & 9 & 7 \end{bmatrix}$$

is lower triangular.

How do we ensure models aren’t memorising grounded concepts?



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Web colors

From Wikipedia, the free encyclopedia

Web colors are colors used in displaying web pages on the World Wide Web, and the methods for describing and specifying those colors. Colors may be specified as an RGB triplet or in hexadecimal format (a *hex triplet*) or according to their common English names in some cases. A color tool or other graphics software is often used to generate color values. In some uses, hexadecimal color codes are specified with notation using a leading number sign (#).[1][2] A color is specified according to the intensity of its red, green and blue components, each represented by eight bits. Thus, there are 24 bits used to specify a web color within the sRGB gamut, and 16,777,216 colors that may be so specified.

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How do we ensure models aren't memorising grounded concepts?

Name	Hex (RGB)	Red (RGB)	Green (RGB)	Blue (RGB)	Hue (HSL/HSV)	Satur. (HSL)	Light (HSL)	Satur. (HSV)	Value (HSV)	CGA number (name); alias
Aqua	#00FFFF	0%	100%	100%	180°	100%	50%	100%	100%	11 (high cyan); cyan
Blue	#0000FF	0%	0%	100%	240°	100%	50%	100%	100%	09 (high blue)
Teal	#008080	0%	50%	50%	180°	100%	25%	100%	50%	03 (low cyan)
Navy	#000080	0%	0%	50%	240°	100%	25%	100%	50%	01 (low blue)
Lime	#00FF00	0%	100%	0%	120°	100%	50%	100%	100%	10 (high green); green
Green	#008000	0%	50%	0%	120°	100%	25%	100%	50%	02 (low green)
Black	#000000	0%	0%	0%	0°	0%	0%	0%	0%	00 (black)
Gray	#808080	50%	50%	50%	0°	0%	50%	0%	50%	08 (dark gray)
Purple	#800080	50%	0%	50%	300°	100%	25%	100%	50%	05 (low magenta)
Olive	#808000	50%	50%	0%	60°	100%	25%	100%	50%	06 (brown)
Maroon	#800000	50%	0%	0%	0°	100%	25%	100%	50%	04 (low red)
Silver	#C0C0C0	75%	75%	75%	0°	0%	75%	0%	75%	07 (light gray)
White	#FFFFFF	100%	100%	100%	0°	0%	100%	0%	100%	15 (white)
Fuchsia	#FF00FF	100%	0%	100%	300°	100%	50%	100%	100%	13 (high magenta);

From all the (training) data on the web, there are many instances of the "groundings" we wish to evaluate (e.g., names of colours) that language models may have seen during training.

How do we ensure models aren’t memorising grounded concepts?

Name	Hex (RGB)	Red (RGB)	Green (RGB)	Blue (RGB)	Hue (HSL/HSV)	Satur. (HSL)	Light (HSL)	Satur. (HSV)	Value (HSV)	CGA number (name); alias
Aqua	#00FFFF	0%	100%	100%	180°	100%	50%	100%	100%	11 (high cyan); cyan
Blue	#0000FF	0%	0%	100%	240°	100%	50%	100%	100%	09 (high blue)
Teal	#008080	0%	50%	50%	180°	100%	25%	100%	50%	03 (low cyan)
Navy	#000080	0%	0%	50%	240°	100%	25%	100%	50%	01 (low blue)
Lime	#00FF00	0%	100%	0%	120°	100%	50%	100%	100%	10 (high green); green
Green	#008000	0%	50%	0%	120°	100%	25%	100%	50%	02 (low green)
Black	#000000	0%	0%	0%	0°	0%	0%	0%	0%	00 (black)
Gray	#808080	50%	50%	50%	0°	0%	50%	0%	50%	08 (dark gray)
Purple	#800080	50%	0%	50%	300°	100%	25%	100%	50%	05 (low magenta)
Olive	#808000	50%	50%	0%	60°	100%	25%	100%	50%	06 (brown)
Maroon	#800000	50%	0%	0%	0°	100%	25%	100%	50%	04 (low red)
Silver	#C0C0C0	75%	75%	75%	0°	0%	75%	0%	75%	07 (light gray)
White	#FFFFFF	100%	100%	100%	0°	0%	100%	0%	100%	15 (white)
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From all the (training) data on the web, there are many instances of the “groundings” we wish to evaluate (e.g., names of colours) that language models may have seen during training.

How do we ensure that we are truly testing grounding and not memorisation?

Isomorphism and Rotated Worlds

Isomorphism and Rotated Worlds

"..systems are capable of understanding if they emulate representations that are isomorphic to underlying meaning under important semantic relations like equivalence."

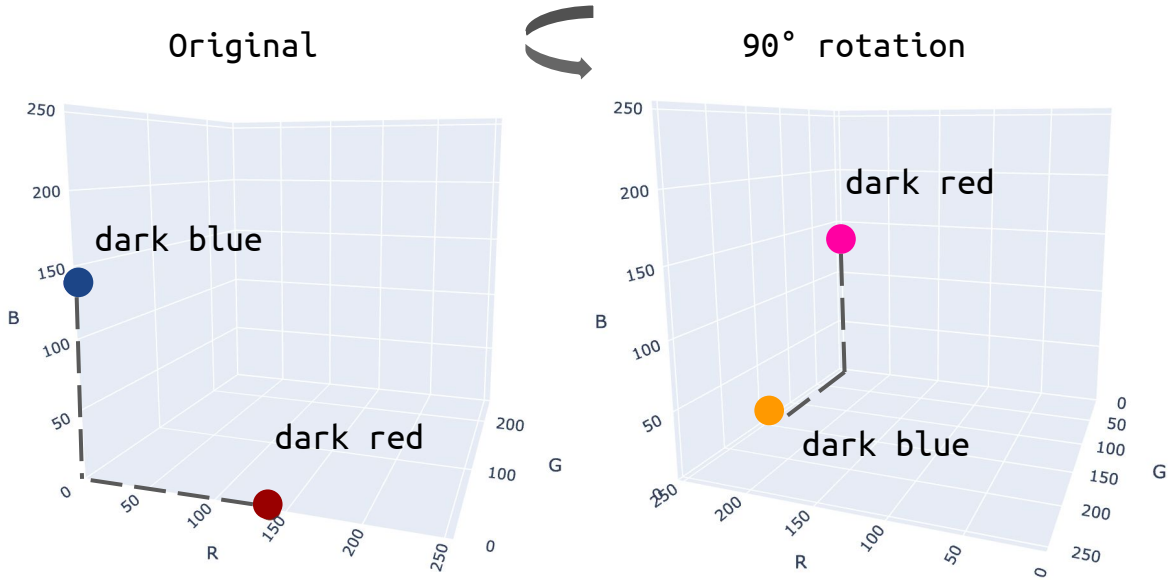
$$f(x) = f(x') \iff g(x) = g(x').$$

Can models perform equally well and use the structure/relation between words under different grounding functions for equivalent worlds?

Provable Limitations of Acquiring Meaning from Ungrounded Form: What Will Future Language Models Understand? Merrill et. al., TACL 2021.

Isomorphism and Rotated Worlds

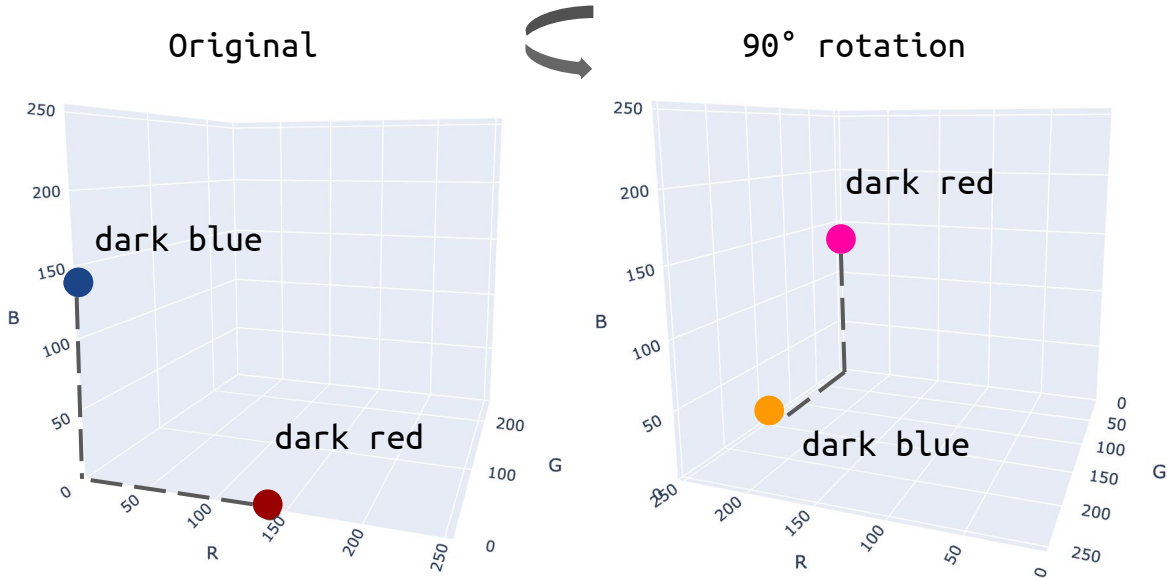
Example: Rotations in colour space



$$f(x') \iff g(x)$$

Isomorphism and Rotated Worlds

Example: Rotations in colour space

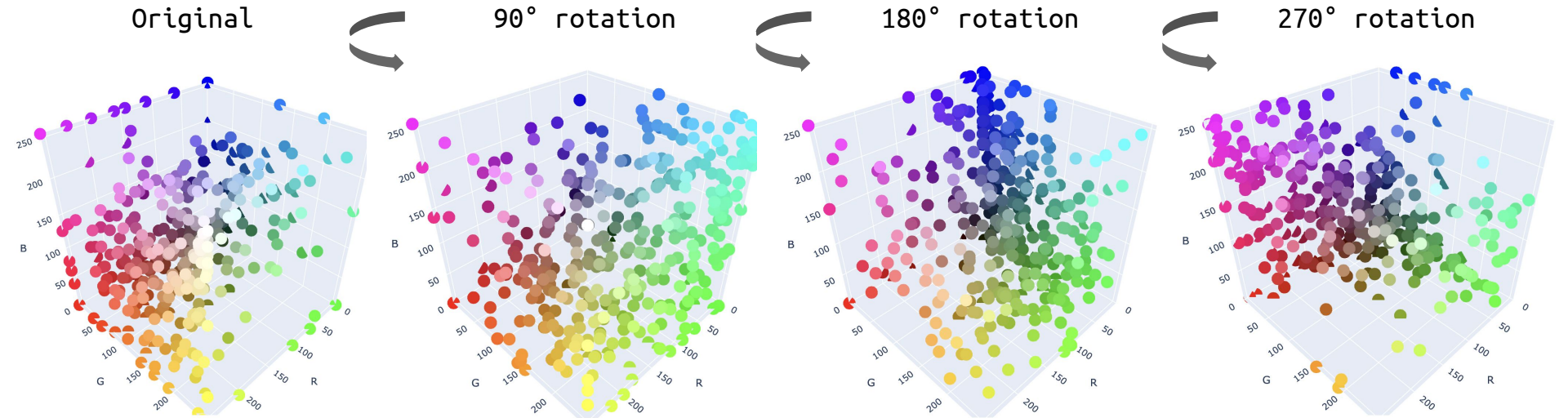


$$f(x') \iff g(x)$$

New "grounding" or semantic meaning of each element in a rotated world

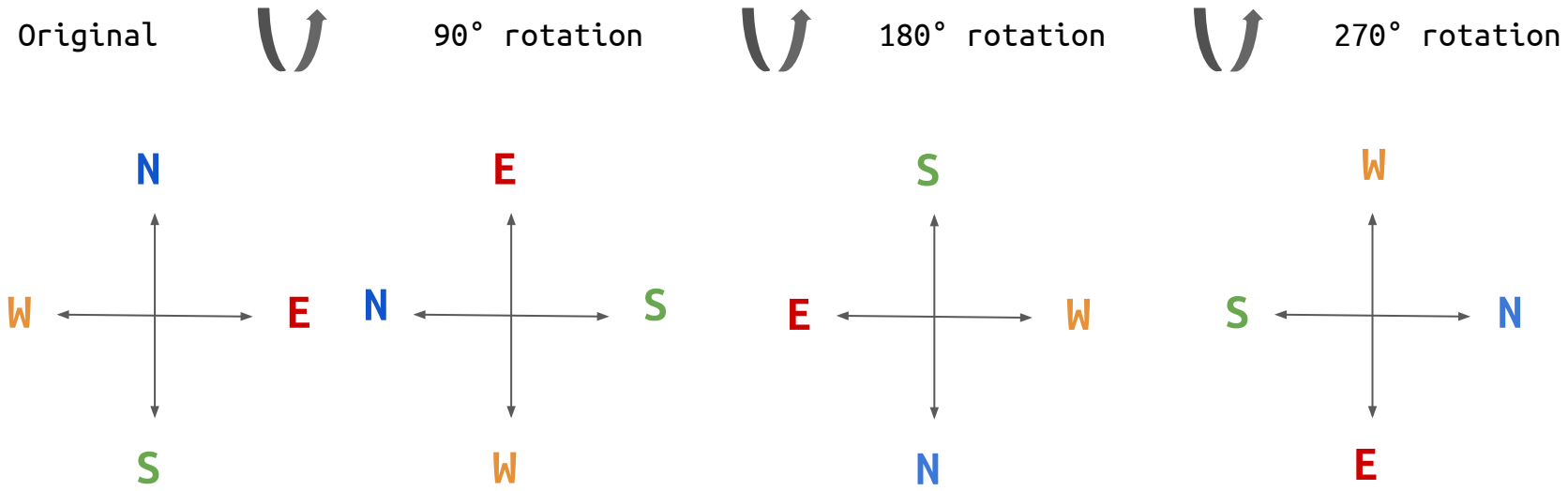
Preserves the structural relation between elements

Rotations in 3D colour space



Rotations in 2D directional space

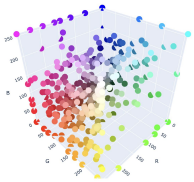
Rotations in cardinal direction space



Isomorphism and Rotated Worlds: Our Expectation

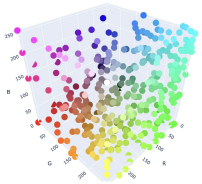
Isomorphism and Rotated Worlds: Our Expectation

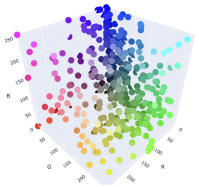
Original

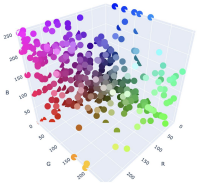




90° or 180° or 270° rotation

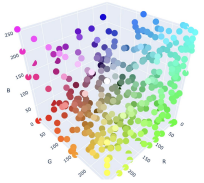






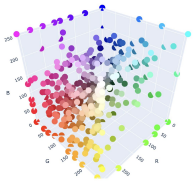


Random scramble

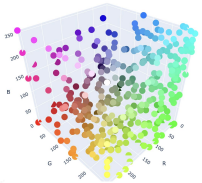


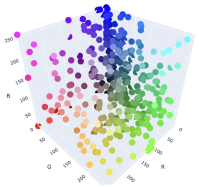
Isomorphism and Rotated Worlds: Our Expectation

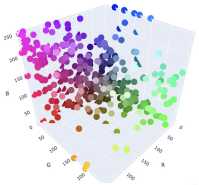
Original



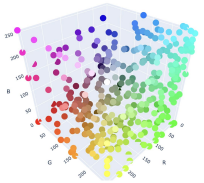
90° or 180° or 270° rotation







Random scramble



$f(x') \iff g(x)$

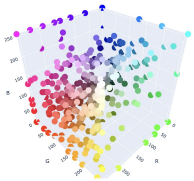
$f(x') \not\iff g(x)$

Preserves relation between word meanings

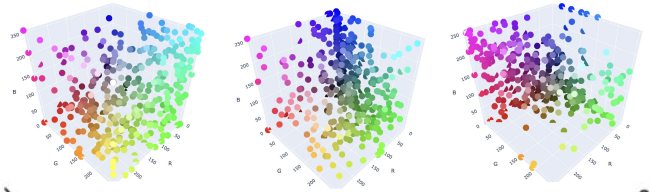
Does not preserve relation between word meanings

Isomorphism and Rotated Worlds: Our Expectation

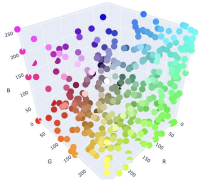
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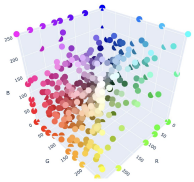
If models know the relation between words..

And can learn a grounding in one world, they should learn a grounding in all worlds that preserve relations.

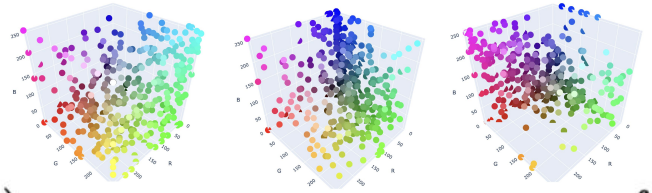
The generalisation pattern is somewhat consistent!

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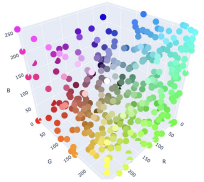
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If models know the relation between words..

And can learn a grounding in one world, they should learn a grounding in all worlds that preserve relations.

The generalisation pattern is somewhat consistent!

A grounding that does not preserve this should be impossible to learn.

The generalisation pattern is random and inconsistent!

Experimental Setup

Experimental Setup

A sanity check (few-shot generalisation)

Can models learn grounded concepts when it is taught to them during (few-shot) training?

Zero-shot generalisation

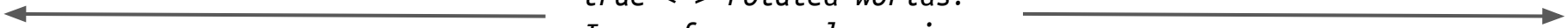
*If we teach a model **some** grounded concepts, can they generalise to other related concepts?*

Compositional generalisation

If we teach a model compositional concepts, can it generalise to unseen compositions?

Error Analysis

How far are model predictions (in the grounded world) from the true grounding?



Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?

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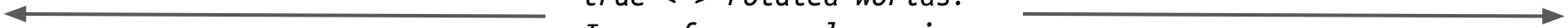
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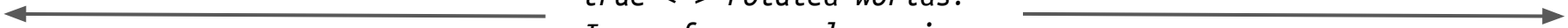
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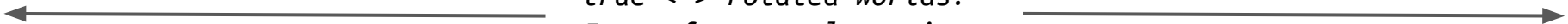
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Metrics

Metrics

Top-1 accuracy: Is the correct answer in the first prediction of the model?

Top-3 accuracy: Is the correct answer in the top three predictions of the model?

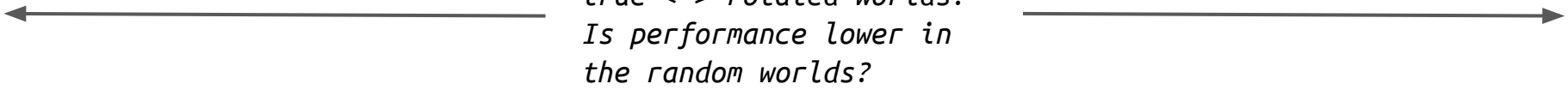
Euclidean distance in the world: How close is the predicted answer from the correct answer (in terms of distance in the world)?

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Top-3 accuracy: Is the correct answer in the top three predictions of the model?

Euclidean distance in the world: How close is the predicted answer from the correct answer (in terms of distance in the world)?



A note on metrics

```
RGB: (0, 0, 255)
Answer: blue

RGB: (255, 0, 0)
Answer: red

RGB: (255, 255, 0)
Answer: yellow

RGB: (255, 165, 0)
Answer: orange

RGB: (143, 0, 255)
Answer: violet

RGB: (0, 255, 0)
Answer: green

RGB: (29, 172, 214)
Answer: bright cerulean

RGB: (223, 0, 255)
Answer: phlox

RGB: (0, 65, 106)
Answer: dark imperial blue

RGB: (111, 0, 255)
Answer: electric indigo

RGB: (239, 222, 205)
Answer:
```

```
Array
[[0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Position:  right

Array
[[0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 1. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]]
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 [0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
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```

Ground-truth answer is usually 1-2 words, but models can autoregressively generate any number of words until the <EOS> token

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Position:  right
```

Ground-truth answer is usually 1-2 words, but models can autoregressively generate any number of words until the <EOS> token

We use a **substring match** metric to compare the models' generation to the ground-truth answer

If the *entire model generation lies in the ground-truth answer*, the model is correct

A note on metrics

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```

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Array
[[0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Position:  right

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 [0. 0. 0.]
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```

Ground-truth answer is usually 1-2 words, but models can autoregressively generate any number of words until the <EOS> token

We use a **substring match** metric to compare the models' generation to the ground-truth answer

If the *entire model generation lies in the ground-truth answer*, the model is correct

This favours **in-domain answers** from the models, or **more general** answers, but discards near-correct samples with other words appended to the answer

Experimental Setup

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Can models learn grounded concepts when it is taught to them during (few-shot) training?

Zero-shot generalisation

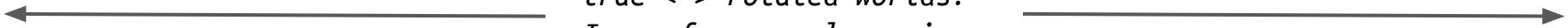
*If we teach a model **some** grounded concepts, can they generalise to other related concepts?*

Compositional generalisation

If we teach a model compositional concepts, can it generalise to unseen compositions?

Distance metrics

How far are model predictions (in the grounded world) from the true grounding?



Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?

Can models learn to ground concepts few-shot?

```
Array
[[0. 0. 0. 0.]
 [0. 0. 0. 1.]
 [0. 0. 0. 0.]]
Position:  right

Array
[[0. 0. 0. 0. 0. 0. 0.]
 [0. 1. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]]
Position:  left

Array
[[0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 1.]]
Position:
```

Each language model gets a "prompt" that contains some number of samples of worlds/concepts

We find that 5-10 samples do not yield good-enough performance, but 20 samples are enough for models to learn concepts (maximum is 50 depending on array sizes/tokens)

*The prompt ends with a **prefix** and a **colon**, after which the language model generates the answer*

*Although LMs might generate upto 100 tokens, we evaluate the **first word** generated and score accuracy over 300 test samples*

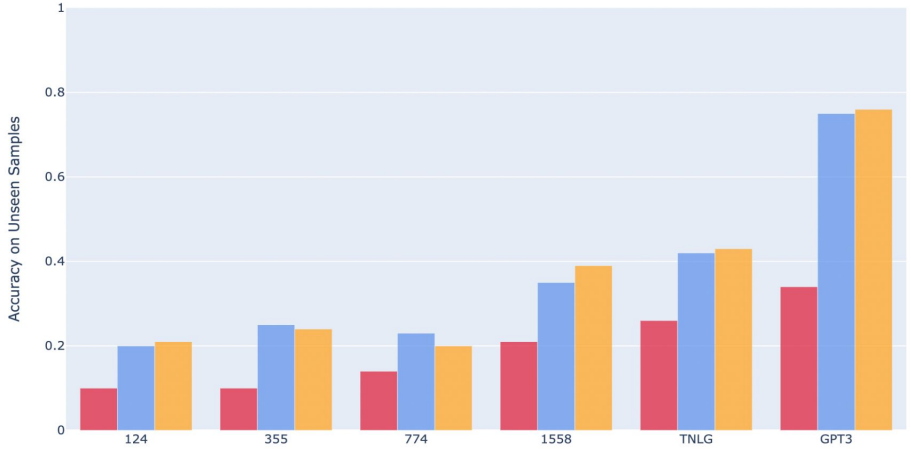
*We look at **top-1** and **top-3** accuracy (i.e., from the most probable words) and evaluate all models on the test set*

Can models learn to ground concepts few-shot?

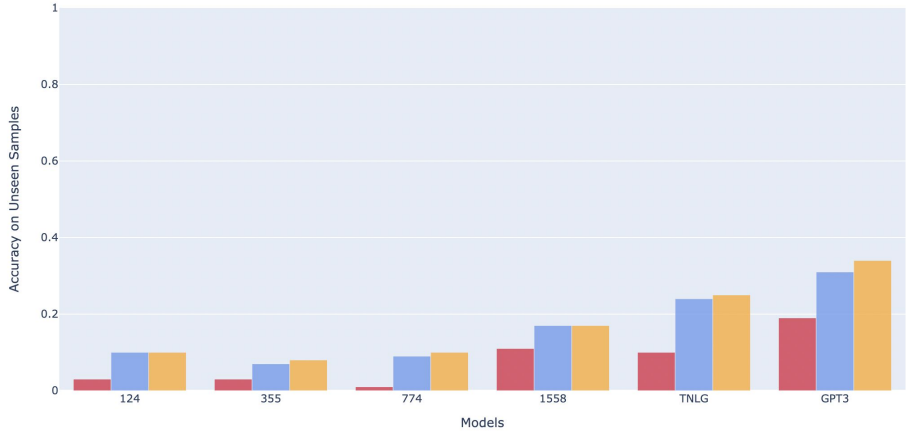
In new unseen worlds, models ***can*** learn to ground what is left vs. right when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right
```

Can models learn to ground concepts few-shot?



Top-3 Accuracy



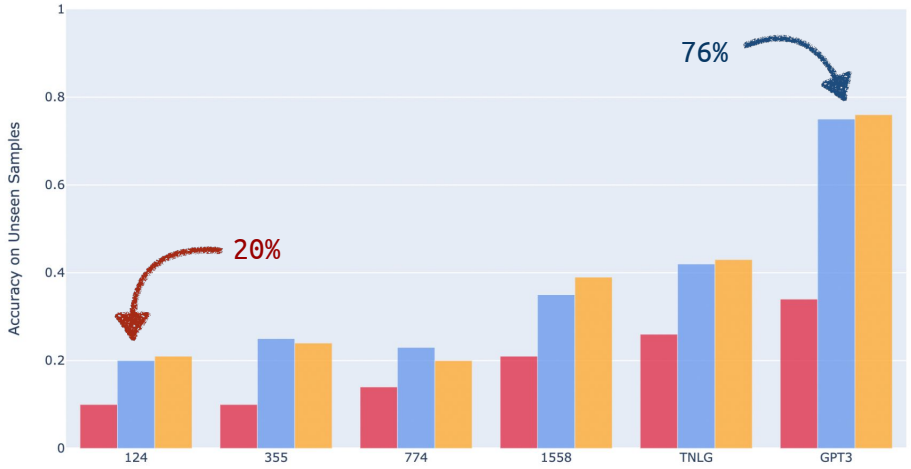
Top-1 Accuracy

Results

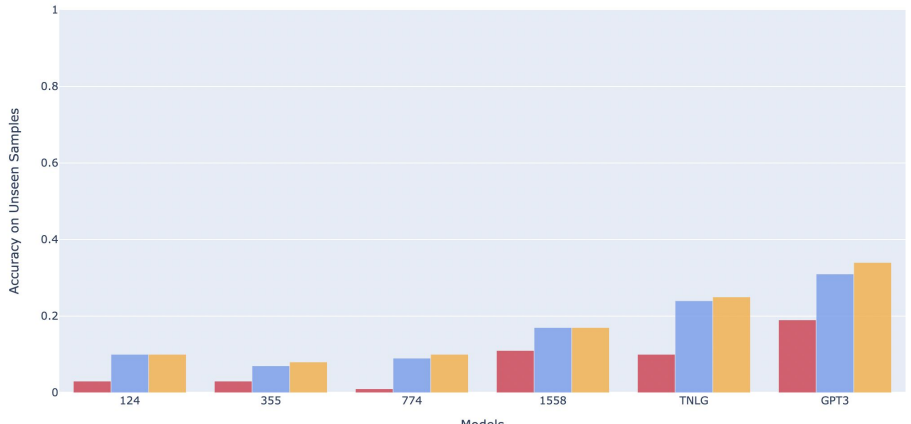
In new unseen worlds, models ***can*** learn to ground what is left vs. right when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right
```

Can models learn to ground concepts few-shot?



Top-3 Accuracy

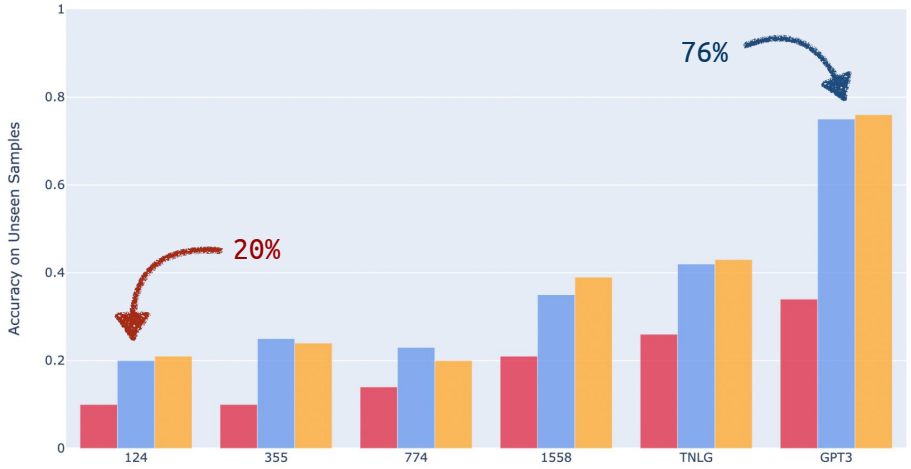


Top-1 Accuracy

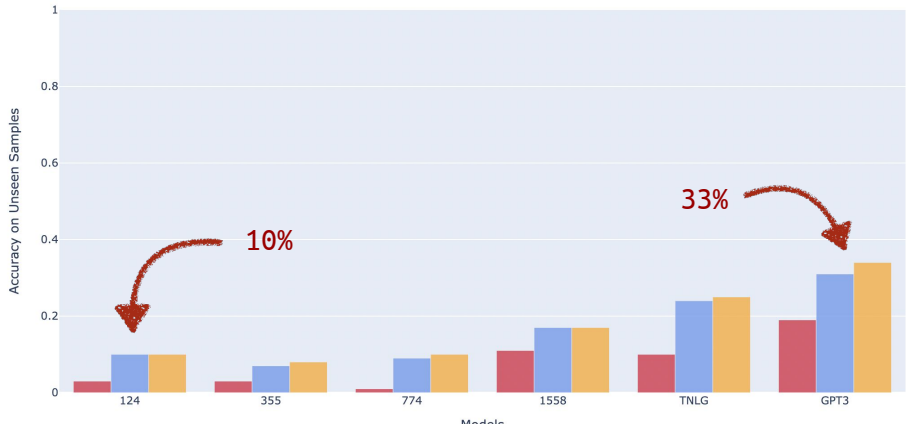
In new unseen worlds, models ***can*** learn to ground what is left vs. right when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right
```

Can models learn to ground concepts few-shot?



Top-3 Accuracy



Top-1 Accuracy

In new unseen worlds, models ***can*** learn to ground what is left vs. right when they have been taught (few-shot) both concepts.

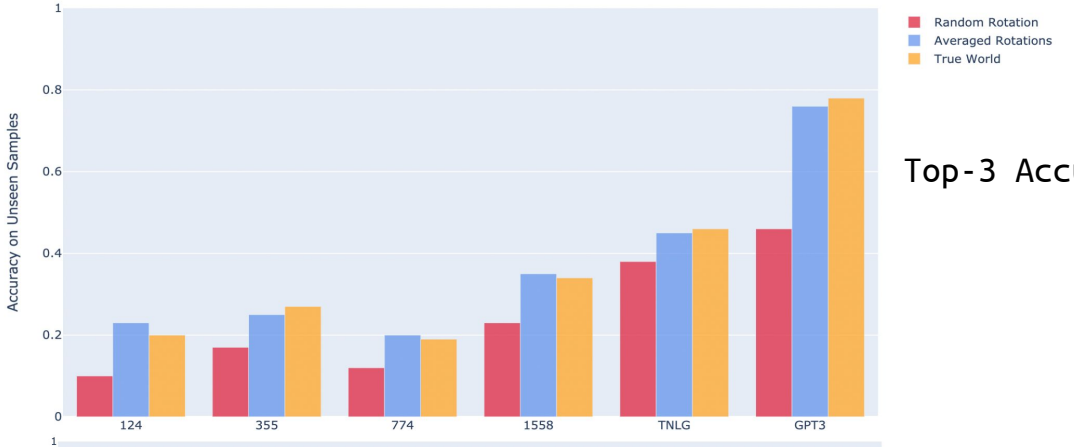
```
Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right
```

Can models learn to ground concepts few-shot?

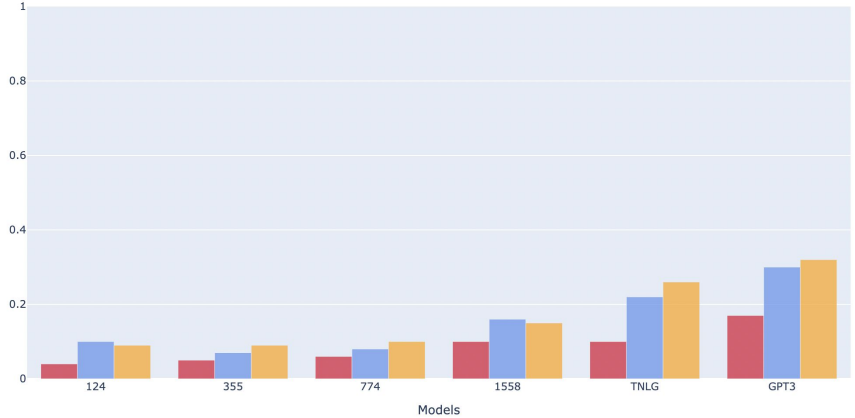
In new unseen worlds, models ***can*** learn to ground what is up vs. down when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 1.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
The direction is:  up
```

Can models learn to ground concepts few-shot?



Top-3 Accuracy



Top-1 Accuracy

In new unseen worlds, models ***can*** learn to ground what is up vs. down when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 1.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
The direction is:  up
```

Can models learn to ground concepts few-shot?



In new unseen worlds, models ***can*** learn to ground what is up vs. down when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 1.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
The direction is:  up
```

Experimental Setup

A sanity check (few-shot generalisation)

Can models learn grounded concepts when it is taught to them during (few-shot) training?

Zero-shot generalisation

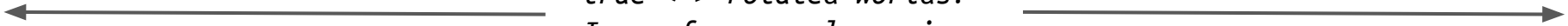
If we teach a model some grounded concepts, can they generalise to other related concepts?

Compositional generalisation

If we teach a model compositional concepts, can it generalise to unseen compositions?

Distance metrics

How far are model predictions (in the grounded world) from the true grounding?



Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?

Can models learn to ground concepts zero-shot?

```
Array
[[0. 0. 1. 0. 0. 0. 0.]]
Position:  left

Array
[[1. 0.]
 [0. 0.]]
Position:  left

Array
[[0. 0.]
 [1. 0.]]
Position:  left

Array
[[1. 0. 0.]
 [0. 0. 0.]]
Position:  left

Array
[[0. 0. 0.]
 [1. 0. 0.]]
Position:  left

Array
[[0. 0. 0. 0.]
 [0. 0. 0. 1.]]
Position:
```

Each language model gets a "prompt" that contains some number of samples of worlds/concepts

We find that 5-10 samples do not yield good-enough performance, but 20 samples are enough for models to learn concepts (maximum is 50 depending on array sizes/tokens)

*The prompt ends with a **prefix** and a **colon**, after which the language model generates the answer*

*Although LMs might generate upto 100 tokens, we evaluate the **first word** generated and score accuracy over 300 test samples*

*We look at **top-1** and **top-3** accuracy (i.e., from the most probable words) and evaluate all models on the test set*

Can models learn to ground concepts zero-shot?

```
Array
[[0. 0. 1. 0. 0. 0. 0.]]
Position:  left

Array
[[1. 0.]
 [0. 0.]]
Position:  left

Array
[[0. 0.]
 [1. 0.]]
Position:  left

Array
[[1. 0. 0.]
 [0. 0. 0.]]
Position:  left

Array
[[0. 0. 0.]
 [1. 0. 0.]]
Position:  left

Array
[[0. 0. 0. 0.]
 [0. 0. 0. 1.]]
Position:
```

Each language model gets a "prompt" that contains some number of samples of worlds/concepts

We find that 5-10 samples do not yield good-enough performance, but 20 samples are enough for models to learn concepts (maximum is 50 depending on array sizes/tokens)

!

*Models **never** see the test concept e.g., "right" but only see related concepts e.g., "left"*

*Although LMs might generate upto 100 tokens, we evaluate the **first word** generated and score accuracy over 300 test samples*

*We look at **top-1** and **top-3** accuracy (i.e., from the most probable words) and evaluate all models on the test set*

Generalising to a random test split

Example Input (60 in-context-learning examples followed by prompt)

RGB: (48, 213, 200)
Answer: orange

RGB: (220, 20, 60)
Answer: crimson

RGB: (143, 0, 255)
Answer:

All 60 Training Examples



6 Primary and Secondary Colours

red, blue, yellow, green, orange, violet

57 Randomly Selected Colours

dark red, navy, electric green, light yellow, fuchsia..

Example Model Outputs

GPT-2 (124M)

color	P=0.14
black	P=0.08
rgb	P=0.02

GPT-3 (175B)

light pink	P=0.24
purple	P=0.13
pink	P=0.08

Train & test splits based on a random partition

Generalising to new sub-spaces

Example Input (60 in-context-learning examples followed by prompt)

RGB: (48, 213, 200)
Answer: orange

RGB: (220, 20, 60)
Answer: crimson

RGB: (0, 0, 128)
Answer:

All 60 Training Examples



6 Primary and Secondary Colours
red, blue, yellow, green, orange, violet

57 Colours Within a Sub-space
dark red, maroon, crimson, fuchsia, rust, bright red..

Example Model Outputs

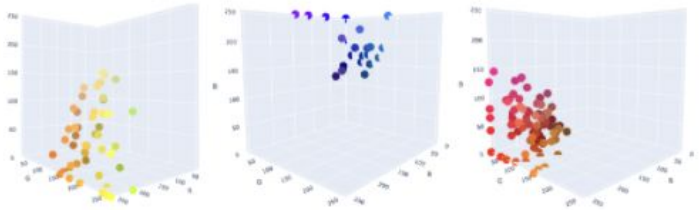
GPT-2 (124M)

color	P=0.14
black	P=0.08
rgb	P=0.02

GPT-3 (175B)

navy	P=0.24
dark blue	P=0.13
blue	P=0.08

Sub-space centered around the colour red



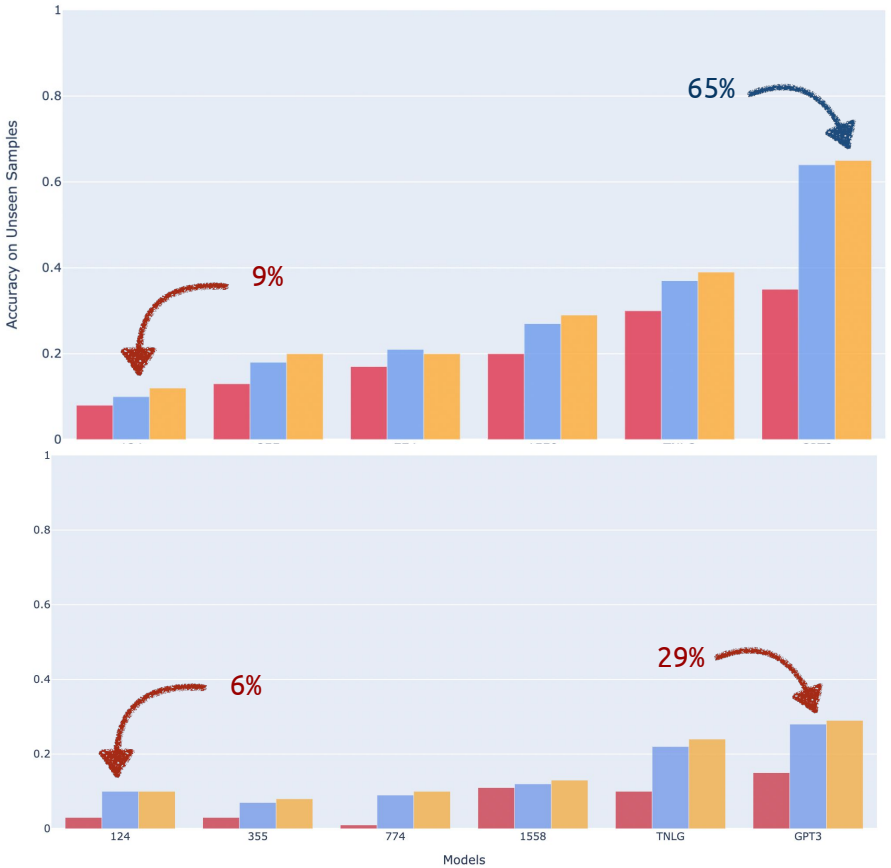
Example colour sub-spaces for yellow, blue and red

Zero-shot generalisation to a random test split

In new unseen worlds, models have to ground what *right* is when they have been taught only *left* (zero-shot).

```
Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right
```

Zero-shot generalisation to a random test split



Top-3 Accuracy

Top-1 Accuracy

In new unseen worlds, models have to ground what *right* is when they have been taught only *left* (zero-shot).

```
Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right
```

Zero-shot generalisation to a random test split

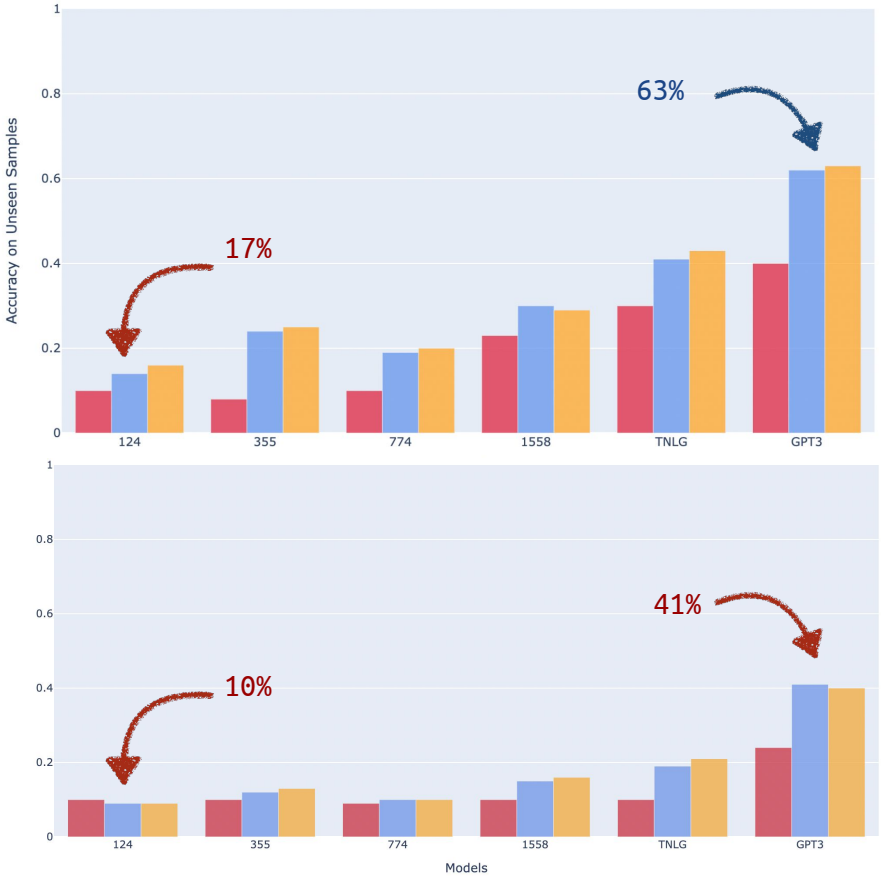
In new unseen worlds, models have to ground what *unseen colours* when they have been taught only *10 colours* (zero-shot).

RGB: (0, 65, 106)
Answer: dark imperial blue

RGB: (111, 0, 255)
Answer: electric indigo

RGB: (239, 222, 205)
Answer:

Zero-shot generalisation to a random test split



Top-3 Accuracy

Top-1 Accuracy

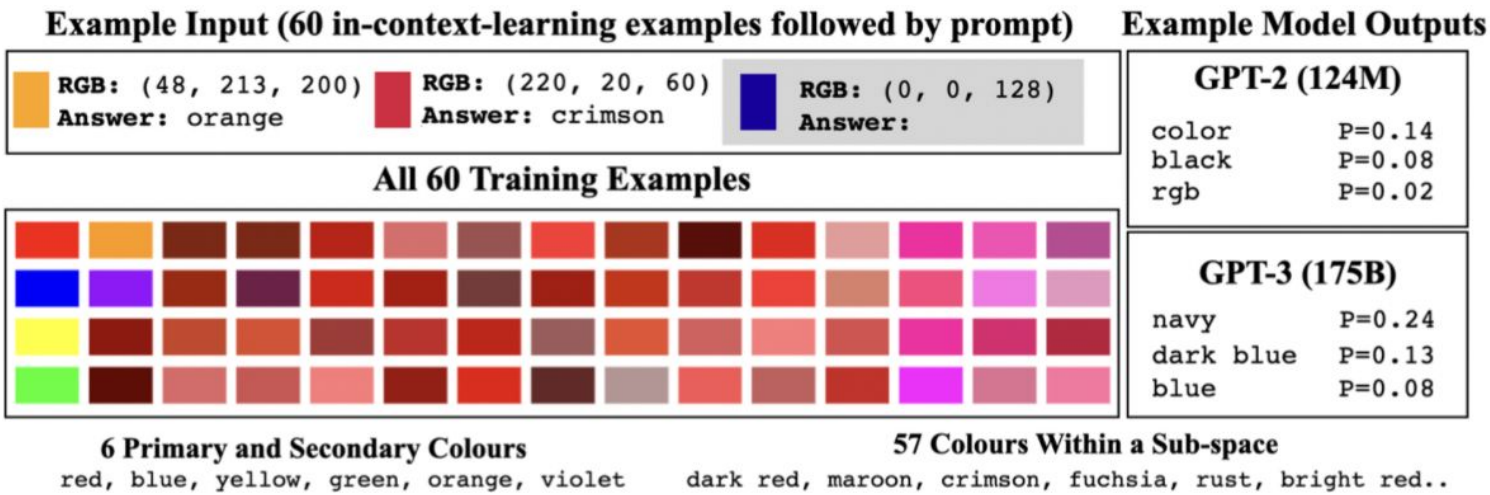
In new unseen worlds, models have to ground what *unseen colours* when they have been taught only *10 colours* (zero-shot).

RGB: (0, 65, 106)
Answer: dark imperial blue

RGB: (111, 0, 255)
Answer: electric indigo

RGB: (239, 222, 205)
Answer:

Zero-shot generalisation to new sub-spaces



Sub-space centered around the colour red

Zero-shot generalisation to new sub-spaces

Example Input (60 in-context-learning examples followed by prompt)

RGB: (48, 213, 200)
Answer: orange

RGB: (220, 20, 60)
Answer: crimson

RGB: (0, 0, 128)
Answer:

All 60 Training Examples



6 Primary and Secondary Colours

red, blue, yellow, green, orange, violet

57 Colours Within a Sub-space

dark red, maroon, crimson, fuchsia, rust, bright red..

Example Model Outputs

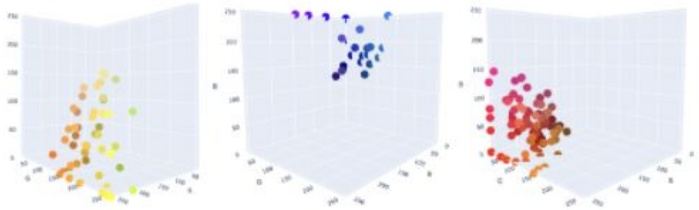
GPT-2 (124M)

color	P=0.14
black	P=0.08
rgb	P=0.02

GPT-3 (175B)

navy	P=0.24
dark blue	P=0.13
blue	P=0.08

Sub-space centered around the colour red



Example colour sub-spaces for yellow, blue and red

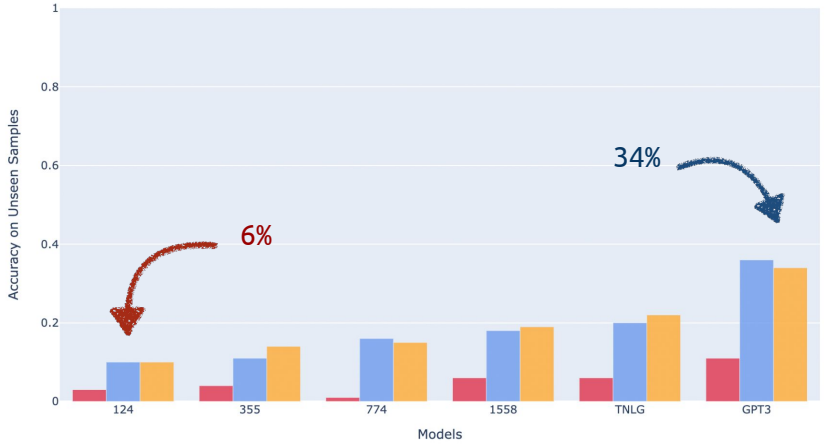
Zero-shot generalisation to new sub-spaces

		Spatial			Cardinal			Colours		
		Original	Rotated	Random	Original	Rotated	Random	Original	Rotated	Random
Top-1 Accuracy	**R-IV**	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	R-ID	0.16	0.16	**0.16**	0.13	0.13	0.13	0.00	0.00	0.00
	124 M	0.10	0.11	0.04	0.11	0.10	0.05	0.08	0.09	0.03
	355 M	0.10	0.10	0.04	0.10	0.11	0.06	0.06	0.07	0.04
	774 M	0.09	0.11	0.03	0.13	0.12	0.08	0.11	0.09	0.01
	1.5 B	0.14	0.14	0.12	0.13	0.14	0.10	0.10	0.09	0.06
	175 B	**0.28**	**0.27**	0.13	**0.30**	**0.29**	**0.08**	**0.23**	**0.21**	**0.11**
Top-3 Accuracy	**R-IV**	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	R-ID	0.16	0.16	0.16	0.13	0.13	0.13	0.00	0.00	0.00
	124 M	0.13	0.12	0.07	0.09	0.09	0.08	0.06	0.05	0.04
	355 M	0.24	0.17	0.15	0.19	0.17	0.10	0.14	0.11	0.12
	774 M	0.19	0.24	0.12	0.17	0.15	0.11	0.15	0.16	0.14
	1.5 B	0.32	0.29	0.20	0.21	0.20	0.14	0.19	0.18	0.16
	175 B	**0.64**	**0.65**	**0.21**	**0.60**	**0.61**	**0.09**	**0.34**	**0.36**	**0.13**

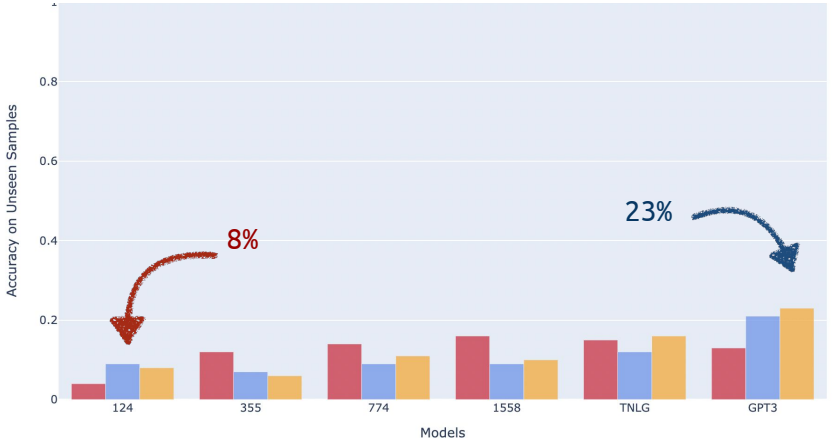
Table 2: Table shows evaluations for generalisation to unseen concepts for the sub-space split of training/test data. Rows show model sizes for GPT-2 models (124M to 1.5B) and the 175B GPT-3 model. Columns show metrics for the original, rotated and random worlds for each of the concept categories. We report both Top-1 and Top-3 accuracy. R-IV refers to a random baseline that randomly selects a word out of the total vocabulary, R-ID refers to a baseline that randomly selects a word out of the set of words in the concept category.

(TODO)

Zero-shot generalisation to new sub-spaces



Top-3 Accuracy



Top-1 Accuracy

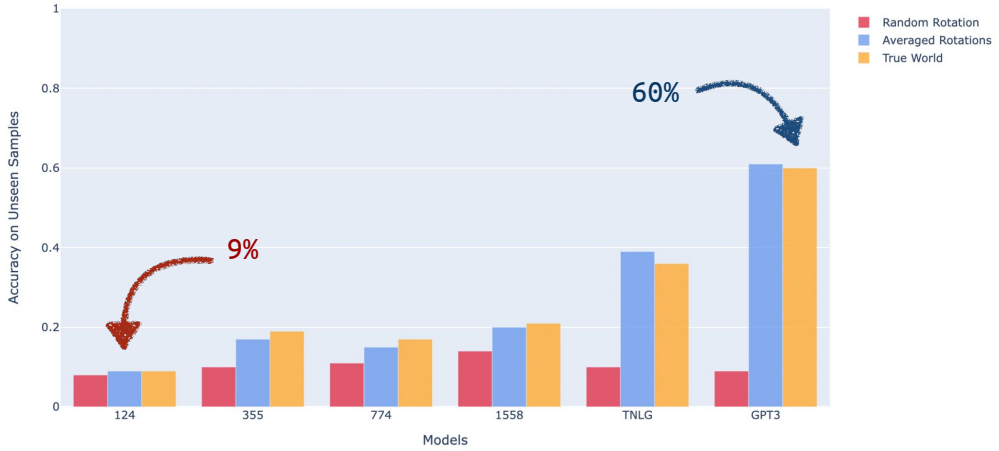
In new unseen worlds, models have to ground what *unseen colours* when they have been taught only *10 colours* (zero-shot).

RGB: (0, 65, 106)
Answer: dark imperial blue

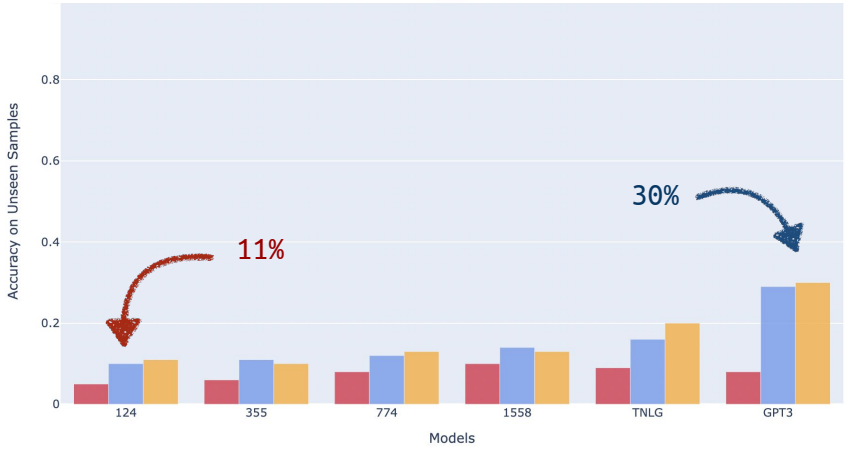
RGB: (111, 0, 255)
Answer: electric indigo

RGB: (239, 222, 205)
Answer:

Zero-shot generalisation to new sub-spaces



Top-3 Accuracy



Top-1 Accuracy

In new unseen worlds, models have to ground what *left* is when they have been taught only *right* (zero-shot).

```
Array
[[0. 0. 0. 0.]
 [0. 1. 0. 0.]]
Position: west
```

Experimental Setup

A sanity check (few-shot generalisation)

Can models learn grounded concepts when it is taught to them during (few-shot) training?

Zero-shot generalisation

*If we teach a model **some** grounded concepts, can they generalise to other related concepts?*

Compositional generalisation

If we teach a model compositional concepts, can it generalise to unseen compositions?

Distance metrics

How far are model predictions (in the grounded world) from the true grounding?

Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?

Analysing Model Predictions

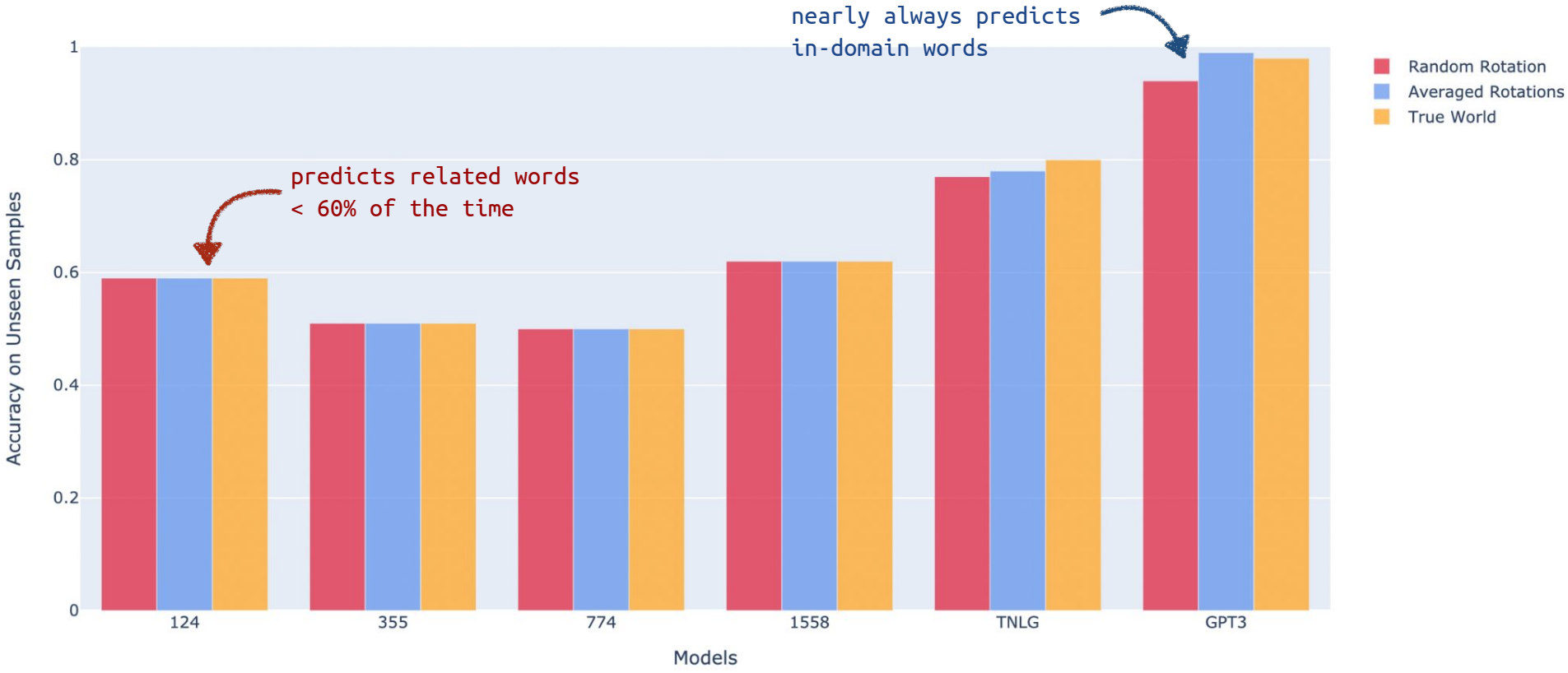
Since models can generate any word from their (10k+) vocabularies, how related are their answers to the concept category?

Relatedness of model predictions

Since models can generate any word from their (10k+) vocabularies, how related are their answers to the concept category?

We score "relatedness" of model predictions, based on whether or not the word is in the set of all possible answers for that concept category (e.g., colours)

Relatedness of model predictions



Relatedness of model predictions: 124M parameter model

Example related generations:

```
magenta
green
aqua
red
orange
purple
yellow
blue
dark blue
tan
silver
cyan
.
.
.
```

Example unrelated generations:

```
depending on your key length (48 bits, no 32
bits), you'll get a variety of brightness
values: red = 0, green = 6.5, blue = 0.5 and
orange = 5 pixels (4 bytes on ordered
hexadecimal).

zeroes

the lossy variety existed a few blemishes in the
hood, irons, axles etc and the tough new moptoid
area correctibility repairbox with drivers and
display was marked as being in good repair. this
recall later proved to be property of the
underhanded mismanaging field operations center
on the back more than 50 years later when…

cape high
```

Relatedness of model predictions: 175B parameter model

Example related generations:

deep magenta
mint
dark chocolate
light purple
light bordeaux
tan
silver
cyan
charcoal
green
peach
deep indigo
.
.
.

Example unrelated generations:

liver
sherbet
rhodolite

Almost no unrelated generations!

Qualitatively analysing distances in predictions

When model predictions are related (but not the one true answer/word) how far in the world are model predicted groundings from the true grounding?

Qualitatively analysing distances in predictions

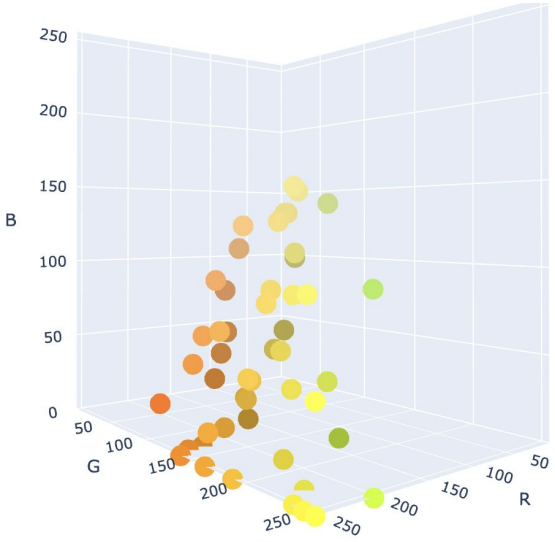
Model Prediction	True Grounding
dark red	wine
medium apricot	apricot
burnt orange	orange
green	forest green
aquamarine	sea green
dark rose	deep carmine
light red	saffron
black	ebony
light purple	lavender
.	.
.	.
.	.

GPT3

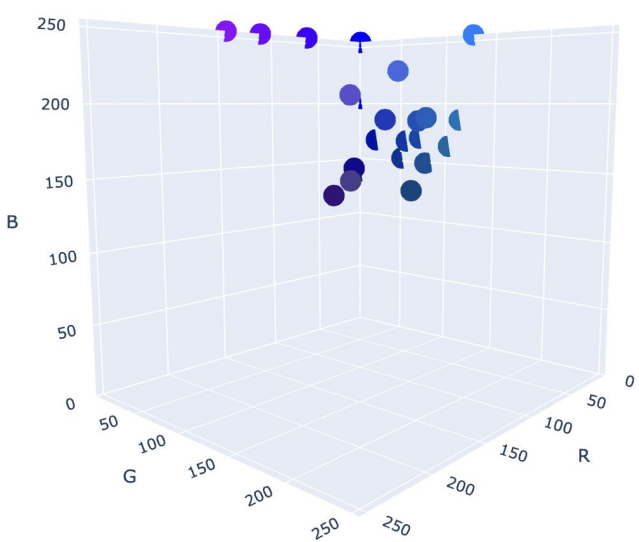
Quantifying distances in predictions

Points within a Euclidean distance of 180 from colours:

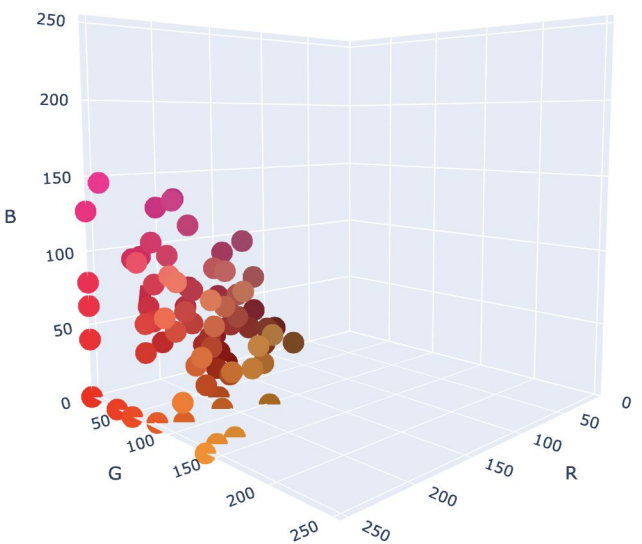
Yellow



Blue



Red



Quantifying distances in predictions

	124M	355M	774M	1.5B	175B
C	328.3	309.5	209.6	190.7	96.3
R-IV	334.9	334.9	334.9	334.9	334.9
R-ID	174.9	174.9	174.9	174.9	174.9

Table 1. Average Euclidean distance between predictions and ground-truth answers.

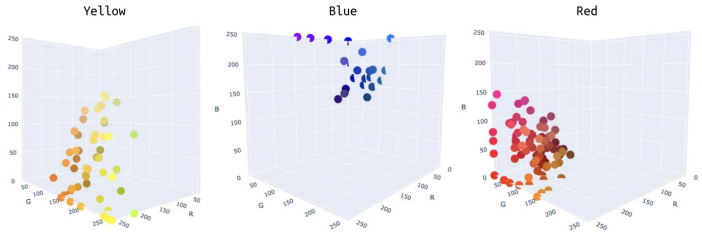
Quantifying distances in predictions

	124M	355M	774M	1.5B	175B
C	328.3	309.5	209.6	190.7	96.3
R-IV	334.9	334.9	334.9	334.9	334.9
R-ID	174.9	174.9	174.9	174.9	174.9

Table 1. Average Euclidean distance between predictions and ground-truth answers.

$$\mathrm{d}(c_1, c_2) = \begin{cases} \sqrt{(c_{1x} - c_{2x})^2 + (c_{1y} - c_{2y})^2 + (c_{1z} - c_{2z})^2} & \text{for} \quad c_1 \in C \\ 500 & \text{for} \quad c_1 \notin C \end{cases}$$

Eq 1. Euclidean distance between two points in space



Quantifying distances in predictions

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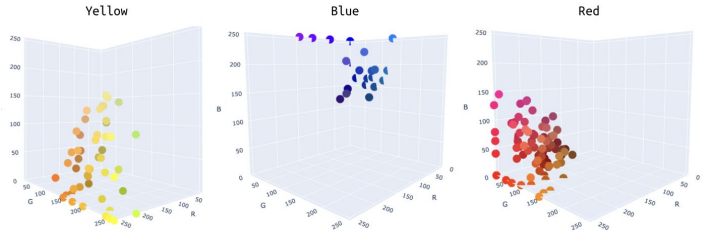
Table 1. Average Euclidean distance between predictions and ground-truth answers.

True G	Predicted G & Distance
dark red	wine (76.5), light crimson (208.1) dark slate gray (144.7)
light green	beige (126.6), light sea green (129.7) cerulean (185.7), violet (262.6)

Table 2. Example predictions from GPT-3 with Euclidean distances from the ground-truth.

$$\mathrm{d}(\mathrm{c}_1, c_2) = \begin{cases} \sqrt{(c_{1x} - c_{2x})^2 + (c_{1y} - c_{2y})^2 + (c_{1z} - c_{2z})^2} & \text{for} \quad c_1 \in C \\ 500 & \text{for} \quad c_1 \notin C \end{cases}$$

Eq 1. Euclidean distance between two points in space



Experimental Setup

A sanity check (few-shot generalisation)

Can models learn grounded concepts when it is taught to them during (few-shot) training?

Zero-shot generalisation

*If we teach a model **some** grounded concepts, can they generalise to other related concepts?*

Compositional generalisation

If we teach a model compositional concepts, can it generalise to unseen compositions?

Distance metrics

How far are model predictions (in the grounded world) from the true grounding?



Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?



Can models learn to ground compositional concepts?

```
RGB: (255, 165, 0)
Answer: orange

RGB: (255, 179, 71)
Answer: pastel orange

RGB: (255, 192, 203)
Answer: pink

RGB: (222, 165, 164)
Answer: pastel pink

RGB: (255, 0, 0)
Answer: red

RGB: (255, 105, 97)
Answer: pastel red

RGB: (143, 0, 255)
Answer: violet

RGB: (203, 153, 201)
Answer: pastel violet
```

Test on blue / pastel blue..

Each language model gets a "prompt" that contains some number of samples of worlds/concepts

We find that 5-10 samples do not yield good-enough performance, but 20 samples are enough for models to learn concepts (maximum is 50 depending on array sizes/tokens)

*The prompt ends with a **prefix** and a **colon**, after which the language model generates the answer*

*Although LMs might generate upto 100 tokens, we evaluate the **first word** generated and score accuracy over 300 test samples*

*We look at **top-1** and **top-3** accuracy (i.e., from the most probable words) and evaluate all models on the test set*

Can models learn to ground compositional concepts?

Each modifier

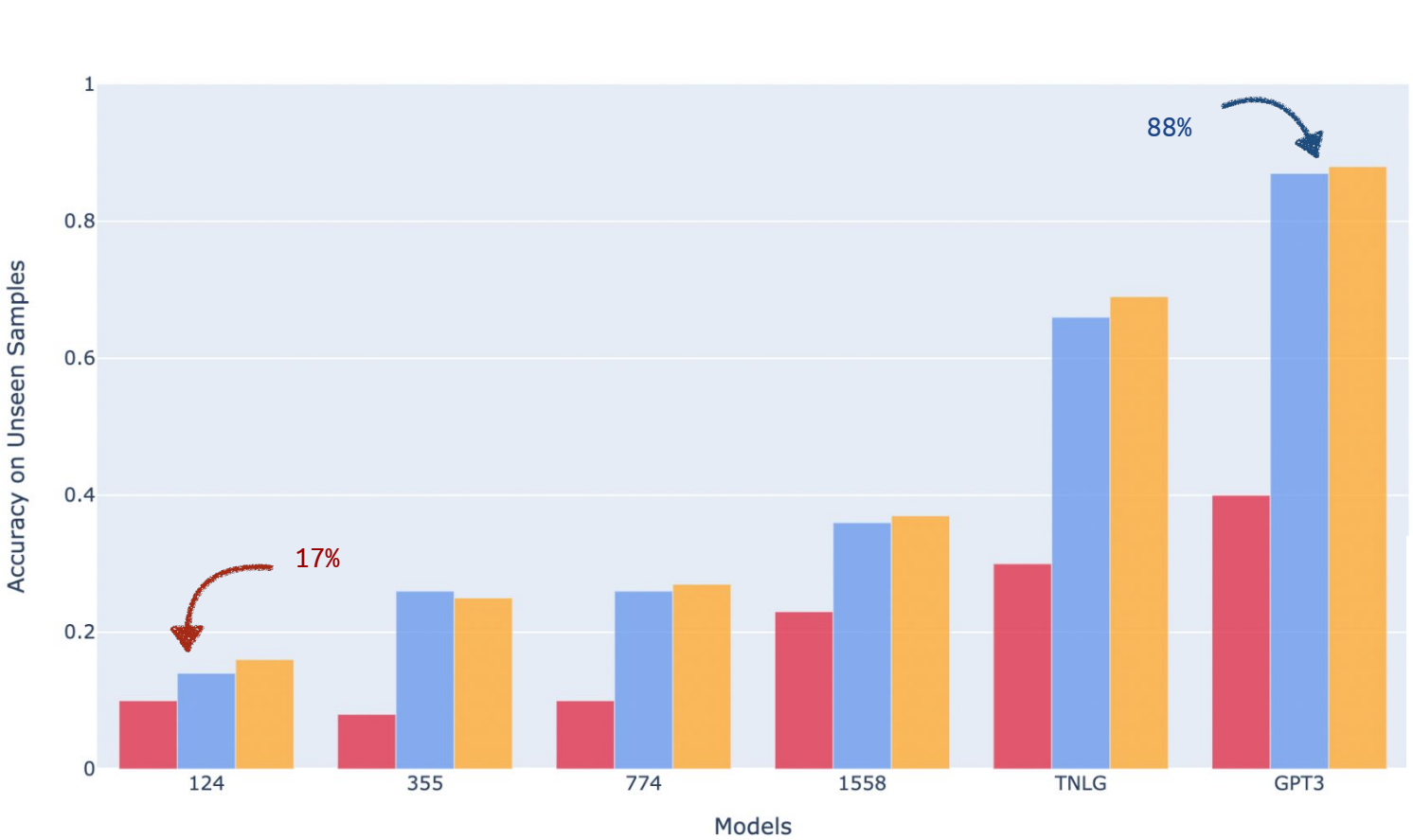
For unseen colours/compositions models have to predict the colour name, given the RGB code

RGB: (143, 0, 255)
Answer: violet

RGB: (203, 153, 201)
Answer: pastel violet

RGB: (255, 255, 0)
Answer:

Can models learn to ground compositional concepts?



For unseen colours/compositions models have to predict the colour name, given the RGB code

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RGB: (203, 153, 201)
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RGB: (255, 255, 0)
Answer:

Can models learn to ground compositional concepts?

All modifiers

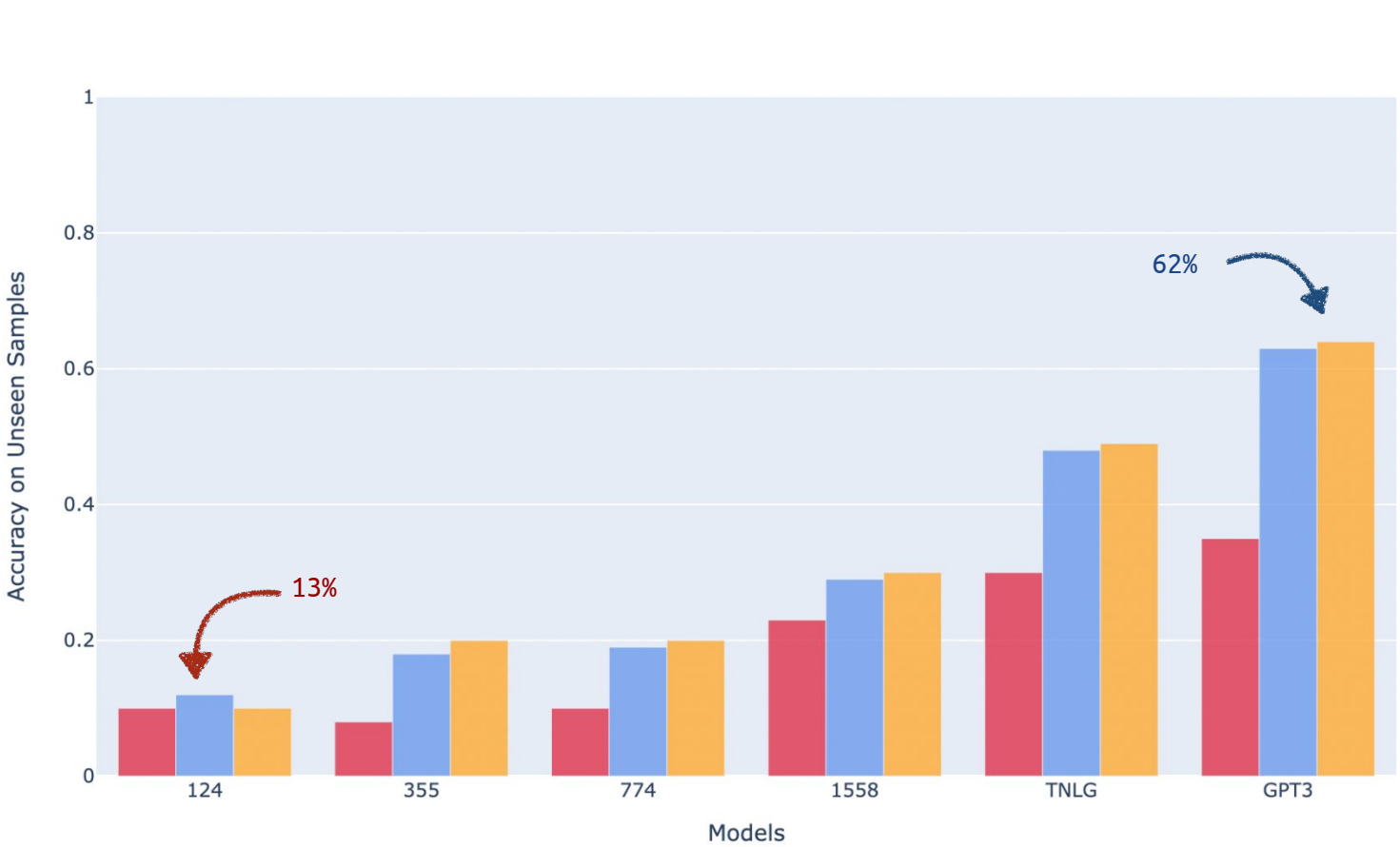
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RGB: (143, 0, 255)
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RGB: (255, 255, 0)
Answer:

Can models learn to ground compositional concepts?



For unseen colours/compositions models have to predict the colour name, given the RGB code

RGB: (143, 0, 255)
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Answer: pastel violet

RGB: (255, 255, 0)
Answer:

In conclusion

In conclusion

A sanity check (few-shot generalisation)	*Zero-shot generalisation*	*Compositional generalisation*	*Distance metrics*
Can models learn grounded concepts when it is taught to them during (few-shot) training?	*If we teach a model **some** grounded concepts, can they generalise to other related concepts?*	*If we teach a model compositional concepts, can it generalise to unseen compositions?*	*How far are model predictions (in the grounded world) from the true grounding?*



Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?



In conclusion

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In conclusion

- *Although large language models have been trained only on text, the structural relations between words give them some amount of information about the grounding of these words in the world*
- *Smaller language models do not seem to learn grounded concepts in few-shot settings, but larger language models can adapt more quickly (still not perfect grounding performance but promising!)*
- *Our work is limited to grounded concepts that can be represented in text (unlike images, 3D worlds..)*
- *Our work shows how we *could* teach language models some elements of the world which has implications for further use-cases in grounded settings*

Thank you!

Questions?

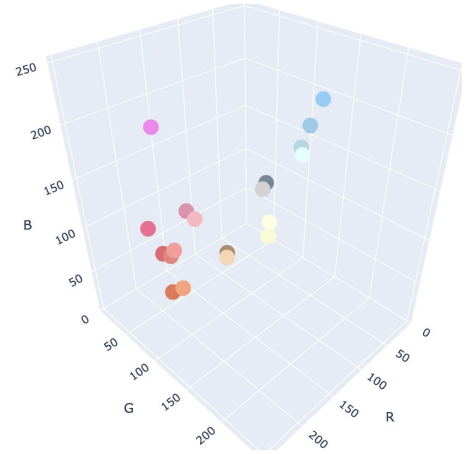
Extra Slides

What happens to modifiers in rotated space?

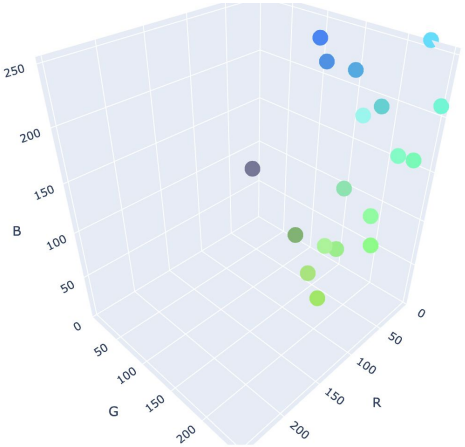
Colour modifiers
(compositions of modifiers and colours)

Average distance = 107.70

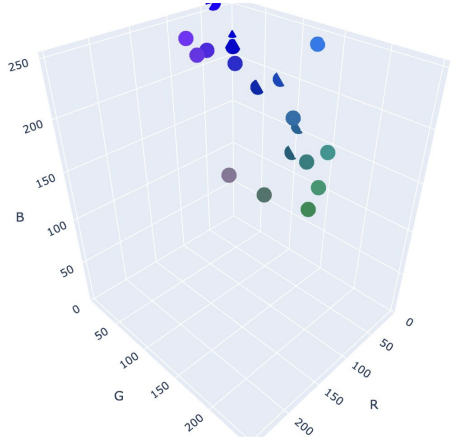
Light



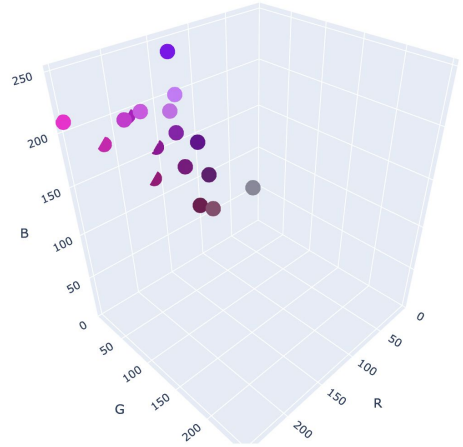
Light (90°)



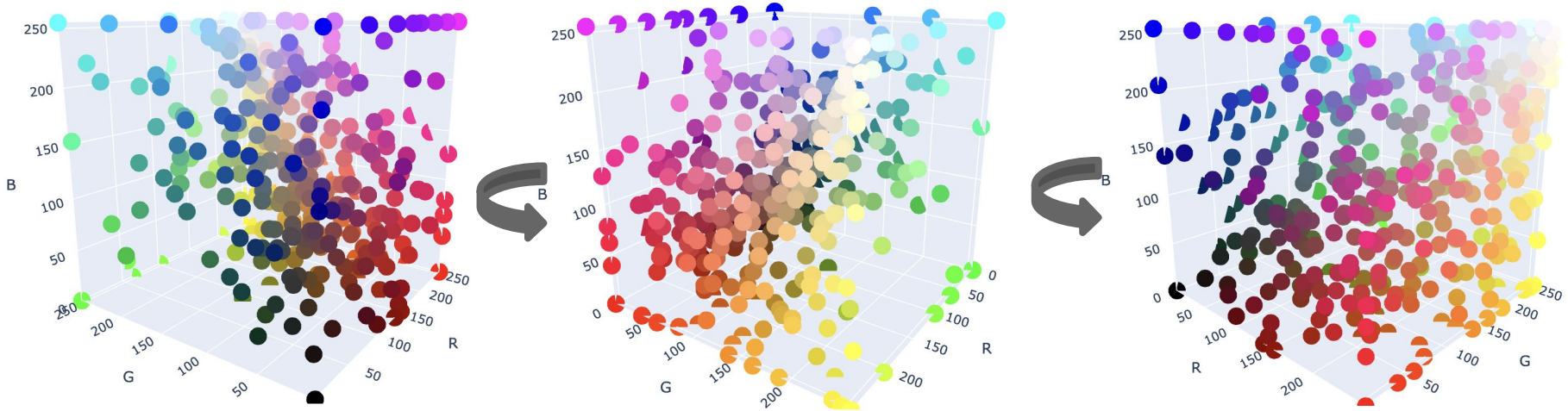
Light (180°)



Light (270°)



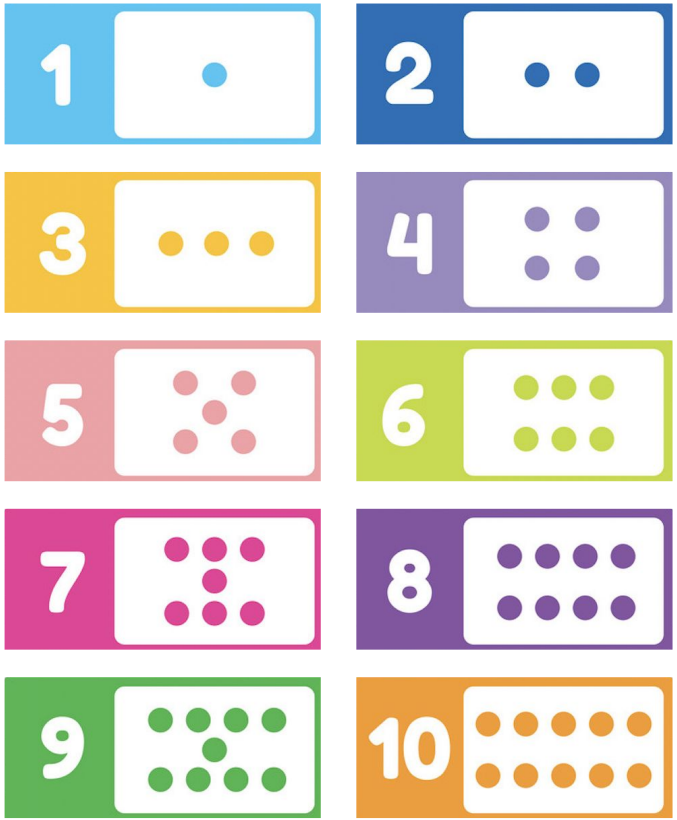
Grounding generative language models



1. For every grounded category we can to test, create rotated worlds/spaces such that the properties of elements within the space remain unchanged

2. Question: are the predictions/representations are isomorphic to the underlying meaning and relation between words for different rotated (equivalent) worlds?

Grounded Categories



Concept: Quantity Terms

If an LM knows all number words e.g., *three, two, one...*

If an LM knows how they relate to each other..

If we show it what it means to have *three* things, can it learn what *two, one, five* mean?

Previous Work

Implicit Representations of Meaning in Neural Language Models

Belinda Z. Li **Maxwell Nye** **Jacob Andreas**
Massachusetts Institute of Technology
{bzl,mnye,jda}@mit.edu

SHAPEWORLD: A new test methodology for multimodal language understanding

Alexander Kuhnle
University of Cambridge
aok25@cam.ac.uk

Ann Copestake
University of Cambridge
aac10@cam.ac.uk

Pay attention to those sets!
Learning quantification from images

doc, I.,[1] Pezzelle, S.,[2] Herbelot, A.,[3] Dimiccoli, M.,[4] Bernardi, R.[5]
[3]*CIMeC - Center for Mind/Brain Sciences,* [5]*CIMeC/DISI, University of Trento*
[4]*CVC - Computer Vision Center, University of Barcelona*
[1,2,3,5]{*firstname.lastname*}*@unitn.it*
[4]{*mariella.dimiccoli*}*@cvc.uab.es*

Previous Work

Prompt Programming for Large Language Models: Beyond the Few-Shot Paradigm

Laria Reynolds
moire@knc.ai

Kyle McDonell
kyle@knc.ai

AUTOPROMPT: Eliciting Knowledge from Language Models with Automatically Generated Prompts

Taylor Shin[*◇] **Yasaman Razeghi**[*◇] **Robert L. Logan IV**[*◇]
Eric Wallace[♠] **Sameer Singh**[◇]
[◇]University of California, Irvine [♠]University of California, Berkeley
{tshin1, yrazeghi, rlogan, sameer}@uci.edu
ericwallace@berkeley.edu

Language Models are Few-Shot Learners

Tom B. Brown[*] **Benjamin Mann**[*] **Nick Ryder**[*] **Melanie Subbiah**[*]

Teaching a Language Model to be Grounded

Climbing towards NLU:
On Meaning, Form, and Understanding in the Age of Data

Emily M. Bender
University of Washington
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Alexander Koller
Saarland University
Dept. of Language Science and Technology
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Provable Limitations of Acquiring Meaning from Ungrounded Form:
What Will Future Language Models Understand?

William Merrill[*] **Yoav Goldberg**[*†] **Roy Schwartz**[‡] **Noah A. Smith**[*§]
[*] Allen Institute for AI [†] Bar Ilan University
[‡] Hebrew University of Jerusalem [§] University of Washington
{willm,yoavg,roys,noah}@allenai.org

Summary of previous work

1. Work that shows language models' progress on text-based **language understanding** tasks
2. Work that outlines why language models cannot have learned **grounded** meaning, and why this is an issue
3. Work that outlines how we could teach language models **new tasks with few-shot prompts** i.e., no fine-tuning to update parameters
4. Work that creates data and tasks to **evaluate grounded language** understanding

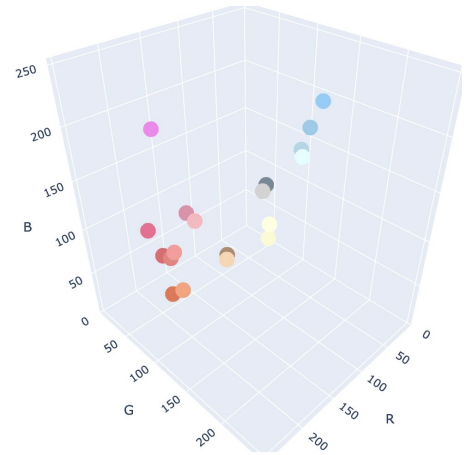
Our work: teach language models a grounded concept in a few-shot manner and see if it can use it's knowledge to generalise to the spectrum of other similar concepts

Modifiers in Rotated Space

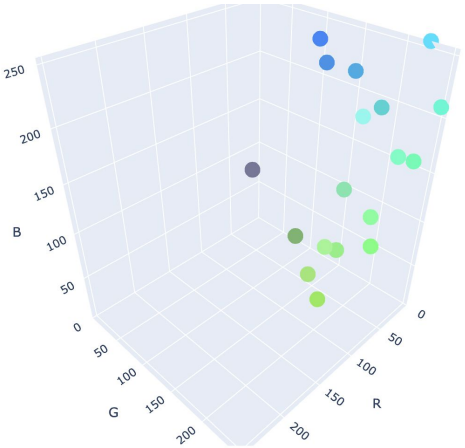
Colour modifiers (compositions)

Average distance = 107.70

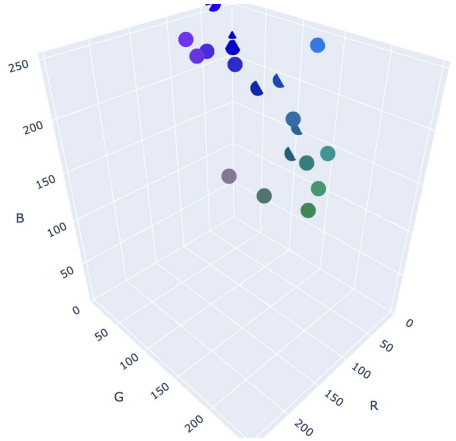
Light



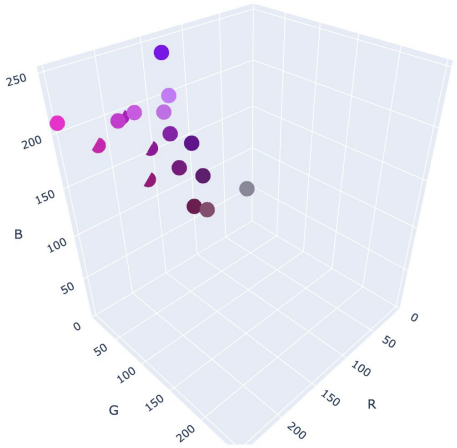
Light (90°)



Light (180°)



Light (270°)

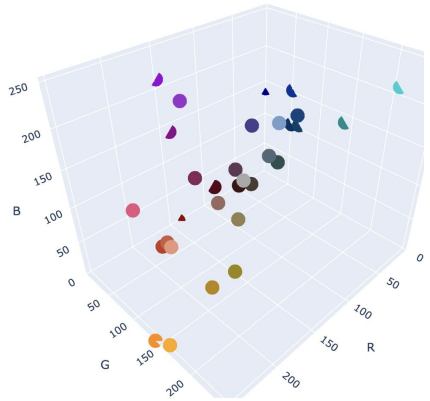


Modifiers in Rotated Space

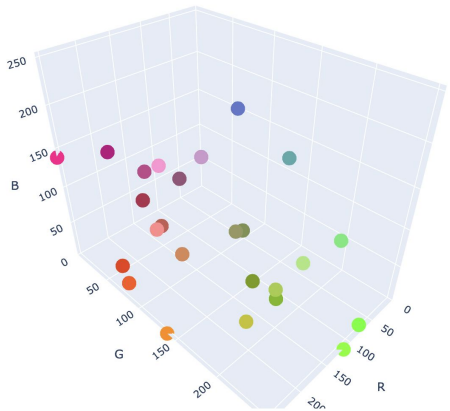
Colour modifiers (compositions)

Average distance = 156.66

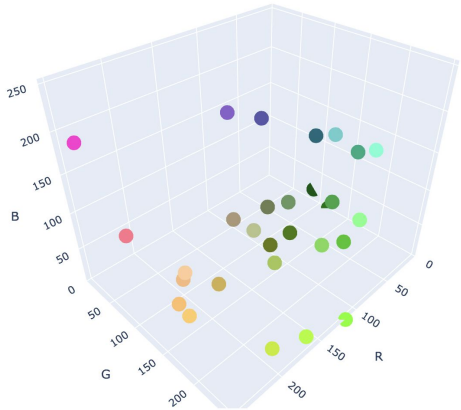
Dark



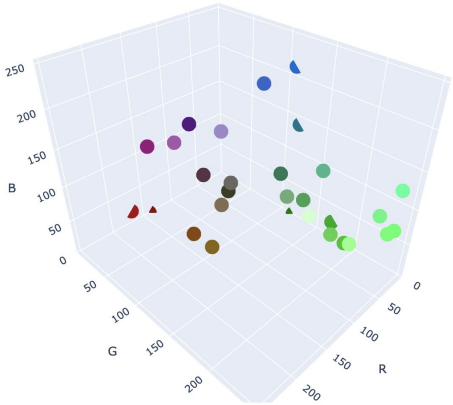
Dark (90°)



Dark (180°)



Dark (270°)

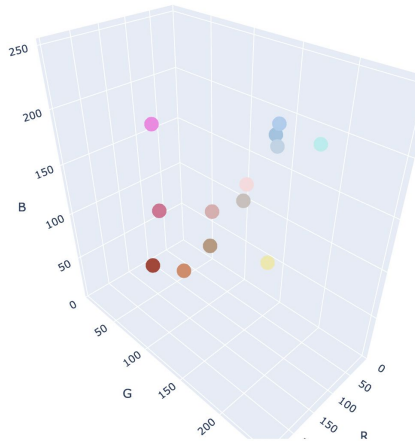


Modifiers in Rotated Space

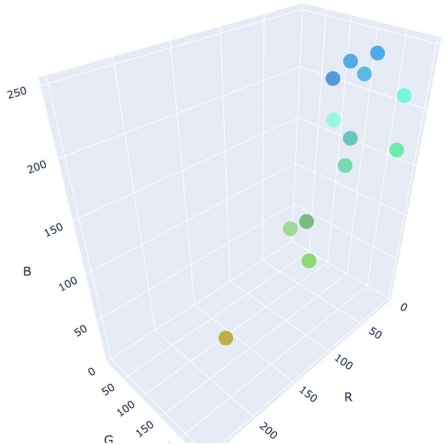
Colour modifiers (compositions)

Average distance = 106.93

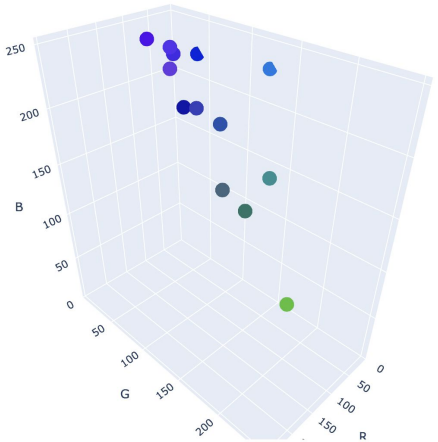
Pale



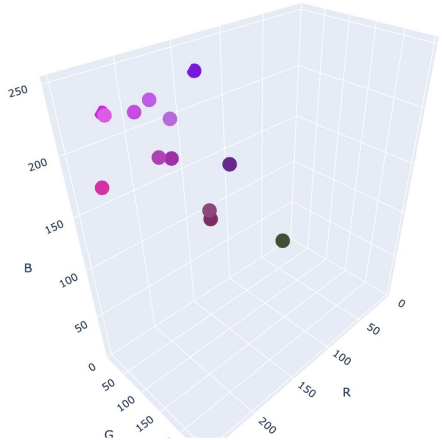
Pale (90°)



Pale (180°)



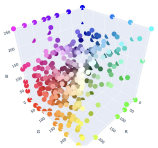
Pale (270°)



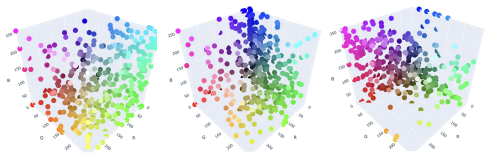
Can models learn to ground concepts?

Question 4: Have models memorised answers (e.g., RGB codes) during training or are they really learning to ground concepts in the world we show them?

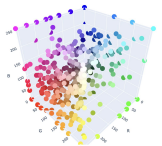
For every true world (e.g., 3D world of RGB coordinates), rotate the world so that the grounded meaning of a term is different, but the elements share the same relation to one another.



True



Rotated



Random

BERT evalutions

		Colour		
		Original	Rotated	Random
	R	0.00	0.00	0.00
	R*	0.13	0.13	0.13
Top-1 Accuracy	**124 M**	0.08	0.09	0.03
	355 M	0.06	0.07	0.04
	774 M	0.11	0.09	0.01
	1.5 B	0.10	0.09	0.06
	175 B	0.23	0.21	0.11
	BERT zero-shot (no-context)	0.09	0.08	0.08
	BERT zero-shot (in-context)	0.08	0.07	0.05
	BERT fine-tuned	0.10	0.09	0.07
	R-IV	0.00	0.00	0.00
	R-ID	0.13	0.13	0.13
Top-3 Accuracy	**124 M**	0.06	0.05	0.04
	355 M	0.14	0.11	0.12
	774 M	0.15	0.16	0.14
	1.5 B	0.19	0.18	0.16
	175 B	0.34	0.36	0.13
	BERT zero-shot (no-context)	0.10	0.09	0.06
	BERT zero-shot (in-context)	0.11	0.09	0.07
	BERT fine-tuned	0.11	0.10	0.09

Table 13: Table shows evaluations for the BERT model when tested in 3 different ways. in the colour domain, where we show models examples of colours in a sub-space (e.g., all colours in a sphere of Euclidean distance 150 from *red*) and test them on different subspaces i.e., all remaining colours. Rows show model sizes for GPT-2 models (124M to 1.5B) and the 175B GPT-3 model. Columns show metrics for the original, rotated and random worlds for each of the concept categories.